

Estimating the Exact Affine Stone Index demand system: the `easi` Stata module

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September 2026

Software: `easi` 2.0.0, <https://github.com/aabdd12/easi>

Zenodo (all versions): <https://doi.org/10.5281/zenodo.22914058>

Abstract

`easi` estimates the Exact Affine Stone Index (EASI) demand system of [Lewbel and Pendakur \(2009\)](#) entirely within Stata and Mata. It is, at the time of writing, the only packaged and maintained implementation of the model in any language: the reference Stata code of [Pendakur \(2008\)](#) is a set of do-files, the R package `easi` was archived by CRAN in 2014, and Stata's official `demandsys` covers the AIDS, QUAIDS and translog families but not EASI. `easi` provides iterated restricted three-stage least squares with the residual covariance iterated to convergence, so that the estimator is invariant to the good dropped; expenditure, uncompensated and compensated price, and demographic elasticities derived from the cost identity, reported as means over the households of the elasticities of each household — those of the household, of the individual or of the market, never those of a reference household; standard errors from the full influence function of each reported mean, which carries the sampling of the households as well as the estimation of the coefficients and of the implicit utility index; sampling weights; robust, cluster, survey-design and bootstrap variances, the survey design read from `svyset` with its finite-population correction; a correction for the households that do not buy a good, by the two-step method of [Shonkwiler and Yen \(1999\)](#) with the implicit utility of the non-buyers completed, and the imputation of their missing prices; `predict`; `estat engel`; `easidiag`, a diagnostic that finds the causes of slow or failed convergence before any estimation is run; and tables with significance stars, exported to Word, L^AT_EX, Excel, CSV or Markdown. Every variance is validated three ways: by the influence function computed by brute force, by a bootstrap of the whole procedure under simple random sampling, sampling weights and a stratified cluster design, and by Monte Carlo experiments on data generated by the model, with and without households that do not buy. An option `compat` reproduces the R package to the eleventh decimal so that published results remain reproducible; [Appendix A](#) records every difference from it with its measured effect.

Keywords: demand systems, EASI, implicit Marshallian demands, elasticities, three-stage least squares, survey design, influence function, censored demand, Shonkwiler–Yen, bootstrap, Monte Carlo.

1 Introduction

Parametric demand systems are convenient but restrictive: Engel curves are constrained by Gorman rank, and unobserved preference heterogeneity is awkward to accommodate. The EASI system of [Lewbel and Pendakur \(2009\)](#) avoids both. Its central device is to replace the unobservable utility level in the Hicksian demands by an *implicit utility index* y that is computable from observed data. The resulting *implicit Marshallian* demands are linear in the parameters given y , admit Engel curves of any shape, and treat the error terms as random utility parameters.

The software landscape is thin. [Pendakur \(2008\)](#) distributes four Stata do-files — iterated 3SLS with and without interactions, and GMM through `gmm` — that estimate the coefficients and stop there: no quantity elasticities, no standard errors on anything but the coefficients, and a comment, twice, that those are wrong for iterated estimates. [Hoareau et al. \(2012\)](#) ported that code to R as the package `easi` and added elasticities, Engel curves, equivalent incomes and simulations; CRAN removed the package on 22 August 2014 and it has not been maintained since. The Stata command `sr_easi` wrapped that package — it exported the data, generated an R script, ran it through `shell`, and read the results back — and so depends on software that can no longer be installed. Stata’s own `demandsys` (Stata 17) fits eight systems, from Cobb–Douglas to the generalised quadratic AIDS, and EASI is not among them. Scattered scripts exist — a set of R functions handling censored shares, conference slides on the `gmm` route, Bayesian estimators in published articles without released code — none packaged.

This note documents `easi`, a self-contained Stata implementation, in enough detail for the estimator, the elasticity formulas, the variance and the diagnostic to be reproduced. Section 2 states the model. Section 3 describes the estimator as implemented, including the numerical treatment of the constrained system. Section 4 derives the elasticities, the three types in which they are reported, and their analytic Jacobian. Section 5 gives the variance estimators — generated regressor, heteroskedasticity, clusters and survey design, elasticities as ratios of estimators — and their validation against the influence function by brute force, a bootstrap of the whole procedure and a Monte Carlo experiment. Section 6 treats the households that do not buy: their missing prices, the censored system, the implicit utility of a non-buyer, the estimator, its variance and its validation. Section 7 describes `easidiag`. Section 8 documents the command, its speed and its results on the reference data. Appendix A records the relation to the R package and the verification of the elasticity formulas; Appendix B lists what the test suite verifies. Every number is produced by the replication scripts of the repository.

This note documents `easi` 2.0.0; the release notes of the repository list what changed from the first version.

2 The EASI demand system

Let there be J goods, indexed $j, k = 1, \dots, J$; T demographic variables z_t ; $\ell_j \equiv \ln p_j$; total expenditure x ; and budget shares w_j . [Lewbel and Pendakur \(2009\)](#) propose the cost function

$$\ln C(p, u, z, \varepsilon) = u + \sum_j m^j(u, z) \ell_j + \frac{1}{2} \sum_j \sum_k a^{jk}(z) \ell_j \ell_k + \frac{1}{2} \sum_j \sum_k b^{jk} \ell_j \ell_k u + \sum_j \varepsilon_j \ell_j, \quad (1)$$

with

$$m^j(u, z) = \sum_{r=1}^R b_r^j u^r + \sum_{t=1}^T g_t^j z_t + \sum_{t=1}^T h_t^j z_t u, \quad a^{jk}(z) = a_0^{jk} + \sum_{t=1}^T a^{jkt} z_t. \quad (2)$$

Shephard’s lemma gives the Hicksian budget shares,

$$w_j = m^j(u, z) + \sum_k a^{jk}(z) \ell_k + \sum_k b^{jk} \ell_k u + \varepsilon_j. \quad (3)$$

2.1 Implicit utility

Setting $\ln x = \ln C$ in (1) and eliminating $\sum_j m^j \ell_j$ with (3) yields a closed form for the utility level in terms of observables only:

$$y = \frac{\ln x - \sum_j w_j \ell_j + \frac{1}{2} \sum_j \sum_k a^{jk}(z) \ell_j \ell_k}{1 - \frac{1}{2} \sum_j \sum_k b^{jk} \ell_j \ell_k}. \quad (4)$$

The numerator is an affine transformation of the Stone index, whence the name. Substituting y for u in (3) gives the implicit Marshallian demand system

$$w_j = \sum_{r=1}^R b_r^j y^r + \sum_t g_t^j z_t + \sum_t h_t^j z_t y + \sum_k a^{jk}(z) \ell_k + \sum_k b^{jk} \ell_k y + \varepsilon_j. \quad (5)$$

Equations (4) and (5) form a fixed point in (w, y) given (p, x, z, ε) . They are linear in the parameters conditional on y , which is what makes linear estimation possible.

2.2 Restrictions

Adding up ($\sum_j w_j = 1$), homogeneity and Slutsky symmetry impose

$$\sum_j b_r^j = \sum_j h_t^j = 0, \quad \sum_j a^{jkt} = \sum_j b^{jk} = 0, \quad \sum_k a^{jk} = \sum_k b^{jk} = 0, \quad a^{jk} = a^{kj}, \quad b^{jk} = b^{kj}. \quad (6)$$

Adding up and homogeneity are imposed by construction: one equation is dropped and prices are normalised, $\text{np}_k \equiv \ell_k - \ell_J$, so that only $J - 1$ equations and $J - 1$ normalised prices enter. The dropped good is recovered from (6). Symmetry is imposed as cross-equation linear constraints.

3 Estimation

3.1 Iterated linear three-stage least squares

Lewbel and Pendakur (2009) note that the exact model can be estimated by GMM or nonlinear 3SLS, and recommend iterated linear 3SLS, a fixed-point estimator in the sense of Dominitz and Sherman (2005). `easi` implements the latter.

All $J - 1$ equations share the same regressor list and the same instrument list. Write X ($n \times k$) for the common design and Z for the instruments; let $\hat{X} = Z(Z'WZ)^{-1}Z'WX$ with $W = \text{diag}(w_i)$ the weights, Y the $n \times (J - 1)$ matrix of dependent shares, and R the matrix of symmetry restrictions. With $A \equiv \hat{X}'W\hat{X}$, the stacked normal matrix is a Kronecker product, which we exploit rather than forming the $n(J - 1)$ stack:

1. **Restricted 2SLS.** Solve the bordered system

$$\begin{pmatrix} I \otimes A & R' \\ R & 0 \end{pmatrix} \begin{pmatrix} \beta \\ \lambda \end{pmatrix} = \begin{pmatrix} \text{vec}(\hat{X}'WY) \\ 0 \end{pmatrix}.$$

2. **Residual covariance.** With $E = Y - XB$ formed from those coefficients, $\hat{\Sigma}_{jk} = e_j' W e_k / (n_{\text{eff}} - k)$. This is `systemfit`'s `geomean` divisor, $\sqrt{(n - k_j)(n - k_k)}$; because the symmetry restrictions are *cross-equation*, they do not reduce any single equation's coefficient count, so $k_j = k$ for all j .
3. **One GLS step.** Replace $I \otimes A$ by $\hat{\Sigma}^{-1} \otimes A$ and the right-hand side by $\text{vec}(\hat{X}'WY\hat{\Sigma}^{-1})$, and solve again.

4. **Variance.** P , the leading $K \times K$ block of the inverse of the bordered matrix, is the conventional covariance estimator; it satisfies $PAP = P$.

Steps 1–4 are one 3SLS fit. The outer loop updates y :

$$y \leftarrow \frac{y^{\text{stone}} + \frac{1}{2} p'A(z)p}{1 - \frac{1}{2} p'Bp}, \quad y^{\text{stone}} = \ln x - \sum_j w_j \ell_j, \quad (7)$$

where $p'A(z)p$ and $p'Bp$ are read off the current coefficients. Iteration stops when the change in the coefficient vector (with interactions) or in y (without) falls below `tolerance()` (default 10^{-6}), or after `iterate()` passes (default 100). A final fit is then run at the converged y .

3.2 The instrument

y depends on the shares, so it is endogenous. The standard device (Pendakur, 2008) is to instrument with $\tilde{y} = \ln x - \sum_j \bar{w}_j \ell_j$, the Stone index deflated by *mean* rather than own shares, and then to refine it. `easi` follows Pendakur’s two-pass scheme:

1. iterate (7) with instruments built from $\tilde{y}$;
2. form, *once*, $\tilde{y}^* = (\tilde{y} + \frac{1}{2} p'A p) / (1 - \frac{1}{2} p'B p)$ together with its powers $(\tilde{y}^*)^r$ and its interactions $\tilde{y}^* z_t$, $\tilde{y}^* \text{np}_k$;
3. iterate again with that instrument held fixed, then fit once more.

The instrument list is $\{z_t\}$, $\{\text{np}_k\}$, $\{\text{np}_k z_t\}$, $\{(\tilde{y}^*)^r\}$, $\{\tilde{y}^* z_t\}$, $\{\tilde{y}^* \text{np}_k\}$ and a constant. The instrument columns tied to y are rebuilt from the current y at each iteration of the first pass, so that the refined instrument entering the second pass is the one the first pass converged to.

3.3 Iterating the residual covariance

`systemfit` and `reg3` both run with `maxiter = 1`: $\hat{\Sigma}$ is formed once from the 2SLS residuals and the GLS step is taken once. The loop the EASI procedure iterates is the one on y , not the one on Σ .

That matters. For a singular demand system the estimator is invariant to which equation is deleted only if it is the maximum-likelihood one (Barten 1969), that is, only if Σ is iterated to convergence. With one step it is not, and the dependence is not negligible: re-estimating with a different good dropped and permuting back, the price elasticities move by up to 0.20 in absolute value — on a choice that no feature of the data dictates.

`easi` therefore iterates Σ to convergence by default. The same comparison then gives 1.2×10^{-7} , the tolerance of the loop rather than any residual dependence. Table 1 also shows a side benefit that serves as an independent check of the adding-up covariance: the good recovered by adding up and the same good estimated directly agree to machine precision, standard error included.

3.4 Conditioning of the constrained system

The bordered matrix of step 1 mixes two scales. Its leading block has entries of order $\hat{\Sigma}^{-1} \otimes A$, which with small budget shares runs to 10^7 and beyond, while the restriction rows are ± 1 : fourteen orders of magnitude inside one matrix, and an LU solve loses all of them. On a three-good cereals system with mean shares of 0.02, $\text{cond}(\hat{\Sigma}) = 1.5$, $\text{cond}(\hat{\Sigma}^{-1} \otimes A) = 343$, and the bordered matrix has condition number 2.7×10^{14} — the solver returns missing. It does not bite on data with shares near 0.1.

Table 1: Invariance to the dropped good. *hixdata*, $J = 9$, $R = 3$, no interactions.

maximum change on permuting the goods	one-step Σ	iterated Σ
expenditure elasticities	2.3×10^{-3}	9.6×10^{-9}
price elasticities	0.203	1.2×10^{-7}
<i>dropped good, recovered vs. directly estimated (iterated Σ)</i>		
spers : elasticity	1.09766	vs. 1.09766
spers : std. error	0.02605	vs. 0.02605

easi scales the restriction rows to the trace of the block, $R \rightarrow cR$ with $c = \text{tr}(\hat{\Sigma}^{-1} \otimes A)/K$, before solving. This changes nothing: the constraint set is the same, the right-hand side of those rows is zero, and the block of the inverse used as the covariance, $P = A^{-1} - A^{-1}R'(RA^{-1}R')^{-1}RA^{-1}$, is invariant to $R \rightarrow cR$. The reproduction of the R package (Appendix A) is unchanged by it.

3.5 Weights

Weights enter every moment: $\hat{X}$, A , $\hat{\Sigma}$ and all the population means used to form elasticities. **aweights** and **pweights** are normalised to sum to n ; **iweights** are used as they stand, with $n_{\text{eff}} = \sum_i w_i$. A unit **pweight** reproduces the unweighted fit, and **pweights** imply **vce(robust)**. With the elasticities of the individuals (Section 4.3) every weight is multiplied by the size of the household, in the whole estimation.

fweights are refused. On household surveys they are almost always sampling weights in disguise — **fw** = **int(pw*10000)** is the usual device for commands that accept no other weight — and a frequency weight counts each household that many times: with that device, the standard errors would come out a hundred times too small. A sampling weight goes in [**pweight**=], whatever its scale.

4 Elasticities

4.1 Semi-elasticities

Pendakur (2008) gives three semi-elasticities, which follow directly from (5):

$$\Psi_{jk} \equiv \left. \frac{\partial w_j}{\partial \ell_k} \right|_y = a^{jk}(z) + b^{jk}y, \quad (8)$$

$$\aleph_j \equiv \frac{\partial w_j}{\partial y} = \sum_{r=1}^R r b_r^j y^{r-1} + \sum_t h_t^j z_t + \sum_k b^{jk} \ell_k, \quad (9)$$

$$\zeta_{jt} \equiv \frac{\partial w_j}{\partial z_t} = g_t^j + h_t^j y + \sum_k a^{jkt} \ell_k. \quad (10)$$

It is convenient to split $\aleph_j = \aleph_j^* + \sum_k b^{jk} \ell_k$ with $\aleph_j^* \equiv \partial m^j / \partial y$. The compensated expenditure and quantity responses are $S = \Psi + ww' - \text{diag}(w)$ and $\Gamma = \text{diag}(w)^{-1}(\Psi + ww')$.

4.2 Quantity elasticities

The reference paper stops there. Quantity elasticities require the response of y itself, and this is where care is needed: (4) expresses y in terms of the shares, and the shares move with prices and expenditure.

The direct route is to differentiate the identity $\ln x = \ln C(p, y, z, \varepsilon)$, which holds at every price vector. From (1),

$$\frac{\partial \ln C}{\partial \ell_k} = w_k \quad (\text{Shephard}), \quad \Phi \equiv \frac{\partial \ln C}{\partial y} = 1 + \sum_j \aleph_j^* \ell_j + \frac{1}{2} \sum_j \sum_k b^{jk} \ell_j \ell_k. \quad (11)$$

Differentiating $\ln x = \ln C$ at fixed x , and then at fixed p , gives immediately

$$\boxed{\frac{\partial y}{\partial \ell_k} = -\frac{w_k}{\Phi}, \quad \frac{\partial y}{\partial \ln x} = \frac{1}{\Phi}.} \quad (12)$$

The sums in Φ run over all J goods, with the coefficients of the dropped good completed by (6); $\Phi = 1$ exactly when all log prices are zero, that is, at the base-period normalisation, and departs from it with the dispersion of relative log prices rather than with their level, since $\sum_k a^{jk} = \sum_k b^{jk} = 0$ makes it invariant to a common shift.

With (12), the quantity $q_j = xw_j/p_j$ gives

$$\eta_j^x = 1 + \frac{\aleph_j}{w_j \Phi}, \quad (13)$$

$$\eta_j^k = -\mathbf{1}\{j = k\} + \frac{1}{w_j} \left(\Psi_{jk} - \frac{\aleph_j w_k}{\Phi} \right). \quad (14)$$

Two internal checks. First, (13)–(14) satisfy the Slutsky decomposition with the compensated matrix Γ above:

$$\eta_j^k = \underbrace{\Gamma_{jk} - \mathbf{1}\{j = k\}}_{\text{compensated}} - w_k \eta_j^x,$$

as substitution confirms. Second, they are invariant to a common rescaling of all prices and expenditure, because Ψ , $\aleph$ and Φ are.

4.3 The mean of the households: three types of elasticities

Each household has its own elasticities, at its own prices, expenditure and demographics. `easi` reports their mean over the households, each elasticity of household h weighted by its budget share of the good and by a weight $\tilde{\omega}_h$ that sets the type:

type	$\tilde{\omega}_h$	the mean of
<code>households</code> (the default)	ω_h	the households
<code>individuals</code> (the default with <code>hysize()</code>)	$\omega_h n_h$	the persons
<code>market</code>	$\omega_h x_h$	the spending: the total demand

with ω_h the sampling weight, n_h the size of the household and x_h its total expenditure. For the expenditure elasticity, with $a_{hj} = \partial w_{hj} / \partial \ln x = \aleph_{hj} / \Phi_h$ from (13),

$$\frac{\sum_h \tilde{\omega}_h w_{hj} \eta_{hj}^x}{\sum_h \tilde{\omega}_h w_{hj}} = 1 + \frac{\sum_h \tilde{\omega}_h a_{hj}}{\sum_h \tilde{\omega}_h w_{hj}},$$

a ratio of two means that never divides by the share of a single household — the plain mean of the household elasticities does, and explodes when some shares are near zero. With the weights $\omega_h x_h$ numerator and denominator are totals of spending, and the ratio is exactly the elasticity of the total demand of the population, whatever the model. For the individuals the whole estimation — coefficients, variance and means — is weighted by $\omega_h n_h$, as the author's `equaids` and `duvm` do; one estimation gives the households and the market together (`e(elast_*)` and `e(elast_*_mkt)`), and `easi`, `elasticities(market)` displays the second after the first.

Not a reference household. The elasticities are not those of a household at the means of prices, expenditure and demographics. Such a household depends on which mean is taken: the log of the mean expenditure and the mean of the log differ by 0.157 on *hixdata*, where the mean of x is 17% above the exponential of the mean of $\ln x$, and by 0.410 on the Mexican survey (51%). The model is nonlinear in y , so the elasticity at the mean is not the mean of the elasticities, and y being implicit, no household has the mean one.

Table 2 shows the three types on the Mexican survey. They differ most for corn, a necessity whose share falls fast with expenditure: its expenditure elasticity is 0.12 for the households, 0.16 for the persons and 0.09 for the market, whose mean gives the most weight to the households that spend most; the composite good, 96% of the budget, hardly moves. The standard errors differ as much: a mean weighted by expenditure rests on fewer households, and the market elasticity of corn has a standard error more than three times that of the households.

Table 2: The three types of elasticities on the reduced Mexican survey (2,477 households, three goods, $R = 3$, `py`, `vce(svy)`); standard errors in parentheses. `individuals` uses the number of household members. Engel and Cournot aggregation hold for every type to 2×10^{-16} .

good	type	expenditure	own price
corn	households	0.1217 (0.0385)	-0.9616 (0.0480)
	individuals	0.1633 (0.0440)	-0.9620 (0.0515)
	market	0.0866 (0.1397)	-1.0170 (0.0813)
wheat	households	0.4758 (0.0374)	-0.7120 (0.0658)
	individuals	0.4501 (0.0407)	-0.6846 (0.0650)
	market	0.4304 (0.0403)	-0.6164 (0.0919)
other	households	1.0276 (0.0013)	-1.0223 (0.0021)
	individuals	1.0294 (0.0016)	-1.0232 (0.0025)
	market	1.0159 (0.0027)	-1.0135 (0.0023)

4.4 What is reported

Writing $\overline{(\cdot)}$ for the mean weighted by $\tilde{\omega}$, $\sum_h \tilde{\omega}_h(\cdot)_h / \sum_h \tilde{\omega}_h$, and $\bar{w}_j$ for the mean share, the reported elasticities are

$$\hat{\eta}_j^x = 1 + \frac{1}{\bar{w}_j} \overline{\aleph_j / \Phi}, \quad (15)$$

$$\hat{\eta}_j^k = -\mathbf{1}\{j = k\} + \frac{1}{\bar{w}_j} \left(\overline{\Psi_{jk}} - \overline{w_k \aleph_j / \Phi} \right), \quad (16)$$

$$\hat{\zeta}_{jt} = \overline{g_t^j + h_t^j y + \sum_k a^{jkt} \ell_k}, \quad (17)$$

$$\hat{\Gamma}_{jk} = \frac{1}{\bar{w}_j} \left(\overline{\Psi_{jk}} + \overline{w_j w_k} \right). \quad (18)$$

These are what `e(elast_exp)`, `e(elast_price_nc)`, `e(elast_demo)` and `e(elast_price_c)` hold (with the suffix `_mkt` for the market). The compensated elasticity (18) is the weighted mean of those of the households, $(\Psi_{hjk} + w_{hj} w_{hk}) / w_{hj} - \mathbf{1}\{j = k\}$, so it carries the mean of the products of the shares. The R package uses the product of the means, $\bar{w}_j \bar{w}_k$, a hybrid that is the elasticity of no household; the two differ by $\text{Cov}(w_j, w_k) / \bar{w}_j$, from 0.015 to 0.066 on the own-price elasticities of *hixdata* and up to 0.052 on the cross-price ones. `e(slutsky)` is likewise the mean of the Slutsky matrices of the households, negative semidefinite whenever each of them is. Both price tables are indexed [good, price]: row j , column k is the response of good j to the price of good k , and own-price elasticities are on the diagonal. The compensated table is $\hat{\Gamma} - I$;

`e(compensated_q)`, kept for compatibility, is $\hat{\Gamma}$ itself, and `e(elast_price)` is the transpose of `e(elast_price_nc)`, `[price, good]`, as the R package returned it.

Because $\sum_j \aleph_j = 0$ and $\sum_j a^{jk} = \sum_j b^{jk} = 0$, the reported elasticities satisfy Engel aggregation $\sum_j \bar{w}_j \hat{\eta}_j^x = 1$ and Cournot aggregation $\sum_j \bar{w}_j \hat{\eta}_j^k = -\bar{w}_k$ exactly, in every sample. `easi` evaluates both after every estimation and stores the residuals in `e(chk_engel)` and `e(chk_cournot)`; `checks` displays them. A residual above rounding error means the elasticity code is wrong on the data at hand, which the user should know about whether or not they asked.

4.5 The analytic Jacobian of the elasticities

An elasticity depends on $\hat{\beta}$ through the per-household quantities it averages, and the influence function of Section 5.6 carries that dependence as $g' \text{IF}_h(\hat{\beta})$, with

$$g = \frac{\partial \hat{\eta}}{\partial \beta} = \frac{\partial \hat{\eta}}{\partial \beta} \Big|_y + \frac{\partial \hat{\eta}}{\partial y} \frac{\partial y}{\partial \beta},$$

the implicit utility index being itself a function of β through $p'A(z)p/2$ and $p'Bp/2$ in (7), with the closed-form derivative of Section 5.3. The term is small, as the generated regressor of Section 5.3 is, but without it the Jacobian is not the derivative of the quantity reported, and the brute force of Section 5.8 finds the gap. The Jacobian of the whole elasticity vector — expenditure, uncompensated price, demographic, and compensated price when requested — is taken at once, so the covariances between equations enter automatically and the dropped good is treated like any other: its elasticity is a function of β through the completion rules, nothing more.

Completed coordinates. The reported quantities (15)–(18) are written in terms of the *completed* coefficient arrays: b_r^j, g_t^j, h_t^j for $j = 1, \dots, J$; b^{jk} for $j, k = 1, \dots, J$; and, for each equation i , the $(1 + T_p) \times J$ array A_i holding the coefficients of np_k (row 1) and of $\text{np}_k z_t$ (row $1 + t$, for the T_p demographics interacted with prices). The free parameter vector β holds the entries with $j \leq J - 1$; the rest are filled in by adding up:

$$\begin{aligned} b_r^J &= -\sum_{j < J} b_r^j, & g_t^J &= -\sum_{j < J} g_t^j, & h_t^J &= -\sum_{j < J} h_t^j, \\ b^{JJ} &= -\sum_{k < J} b^{jk}, & b^{Jk} &= b^{kJ}, & b^{JJ} &= -\sum_{k < J} b^{Jk}, \\ A_J[t, k] &= -\sum_{i < J} A_i[t, k], & A_i[t, J] &= A_J[t, i], & A_J[t, J] &= -\sum_{i < J} A_i[t, J]. \end{aligned}$$

The completion is a linear map, $\beta \mapsto C\beta$, and C is a sparse $0/\pm 1$ matrix. Its columns are worth writing down, because two of the completions are not symmetric and a gradient derived by hand in free coordinates goes wrong precisely there. A free $y \times p$ entry b^{qi} ($q, i < J$) appears in the completed array at (q, i) with $+1$, at (q, J) and (J, q) with -1 , and at (J, J) with $+1$; a free price entry $A_i[t, q]$ appears at (i, q) with $+1$, at (J, q) and at (q, J) — equation q , price J — with -1 , and at (J, J) with $+1$; a free b_r^j, g_t^j or h_t^j appears at j with $+1$ and at J with -1 . Consequently

$$\frac{\partial \Phi}{\partial b^{qi}} = \frac{1}{2}(\ell_q \ell_i - 2\ell_q \ell_J + \ell_J^2) = \frac{1}{2} \text{np}_q \text{np}_i + \frac{1}{2} \ell_J (\text{np}_i - \text{np}_q),$$

and not $\frac{1}{2} \text{np}_q \text{np}_i$. The second piece is antisymmetric in (q, i) : it lies along the Slutsky-symmetry constraints, so it cannot change $g' \hat{V} g$ when $\hat{V}$ is the restricted covariance, but the entry-wise Jacobian must carry it to be the derivative of the function actually computed.

Derivatives in completed coordinates. Every reported elasticity is a weighted mean of a per-household expression, and every per-household expression is either linear in the completed

coefficients or the ratio of two linear forms, $\aleph_j/\Phi$. Let $Y_r \equiv r y^{r-1}$ and write $\overline{uv}$ for the weighted mean of a product. For the expenditure elasticity of good i ,

$$\begin{aligned}\frac{\partial \hat{\eta}_i^x}{\partial b_r^{i'}} &= \frac{1}{\bar{w}_i} \left[\mathbf{1}\{i = i'\} \overline{Y_r/\Phi} - \overline{\aleph_i Y_r \ell_{i'}/\Phi^2} \right], & \frac{\partial \hat{\eta}_i^x}{\partial h_t^{i'}} &= \frac{1}{\bar{w}_i} \left[\mathbf{1}\{i = i'\} \overline{z_t/\Phi} - \overline{\aleph_i z_t \ell_{i'}/\Phi^2} \right], \\ \frac{\partial \hat{\eta}_i^x}{\partial b^{ac}} &= \frac{1}{\bar{w}_i} \left[\mathbf{1}\{i = c\} \overline{\ell_a/\Phi} - \frac{1}{2} \overline{\aleph_i \ell_a \ell_c/\Phi^2} \right].\end{aligned}$$

For the uncompensated price elasticity of good i with respect to price q , the same three blocks appear with the household share w_q inside the mean and the opposite sign,

$$\frac{\partial \hat{\eta}_i^q}{\partial b_r^{i'}} = \frac{1}{\bar{w}_i} \left[-\mathbf{1}\{i = i'\} \overline{w_q Y_r/\Phi} + \overline{w_q \aleph_i Y_r \ell_{i'}/\Phi^2} \right],$$

and likewise for h and b^{ac} , plus the derivative of $\overline{\Psi_{iq}}$: $\overline{z_t}$ on $A_i[1+t, q]$ (with $z_0 \equiv 1$ for the constant row) and $\bar{y}$ on b^{qi} . The demographic elasticity is linear: 1 on g_t^i , $\bar{y}$ on h_t^i , $\bar{\ell}_k$ on $A_i[1+t, k]$. The compensated table is $\overline{z_t}/\bar{w}_q$ on $A_i[1+t, q]$ and $\bar{y}/\bar{w}_q$ on b^{qi} . Each block is one weighted cross-product of an $n \times R$, $n \times T$ or $n \times J$ matrix with an $n \times J$ one; the whole Jacobian in completed coordinates, G_c , costs of the order of nJ^4 operations, and

$$G = G_c C$$

is the Jacobian with respect to β , which multiplies the influence function of the coefficients. On the reference specification of Section 8 (576 parameters, 216 elasticities) this adds nothing measurable to the running time; against forward differences on the reported quantities it agrees to the finite-difference noise floor, a few 10^{-6} on the two nonlinear families and 10^{-11} on the two linear ones, on every specification in the test suite, weighted and survey-designed included.

Section 5.6 adds to every elasticity the sampling term of the two means it is the ratio of; the rows of G above are what that derivation starts from.

5 Variance estimation and its validation

5.1 The bootstrap used throughout

Every variance **easi** reports is validated against a bootstrap of the *whole* iterated procedure — instruments, both passes, the Σ loop, the elasticities — with 500 replications, whose own Monte Carlo error is about 3% of each standard error. The bootstrap resamples the sampling unit of the design in force, and the weights travel with the rows: households under simple random sampling and under **pweights**; primary sampling units drawn with replacement within strata, $m_h = n_h$, under a survey design. The last point is what makes the bootstrap an oracle rather than a mirror: a bootstrap that resampled households would share the estimator's omission of the clustering and could not detect it. (Stata's **bootstrap** refuses weights on the prefixed command; the test suite wraps **easi** in an **eclass** program that carries them.) Two data sets are used: *hixdata*, the reference data of Pendakur (2008), unweighted, and a reduced Mexican expenditure survey (ENIGH 2014, cereals) with 2,477 households, 703 PSUs and 28 strata, built so that every stratum keeps all its PSUs and the design is declared with **svyset**. Ratios below are the reported standard error divided by the bootstrap one unless stated otherwise. Two further instruments close the section: the influence function computed by brute force (Section 5.8), which checks the formula without any sampling noise, and a Monte Carlo experiment on data generated by the model (Section 5.9), which checks it against the dispersion it is meant to estimate.

5.2 The sandwich

Because $\hat{\beta} = Pm$ is linear in the moment vector $m = \hat{X}'W\hat{\Sigma}^{-1}y$, the sandwich estimator is $\widehat{\text{Var}}(\hat{\beta}) = P\hat{B}P$ with $\hat{B} = \sum_i s_i s_i'$ and, for observation i , $s_i = w_i(u_i \otimes \hat{x}_i')$, $u_i = \hat{\Sigma}^{-1}e_i'$. The

identity $PAP = P$ guarantees that this reduces to P under correct specification, so the same expression serves the restricted and unrestricted cases. `vce(robust)` applies $n/(n-1)$ and `vce(cluster)` sums s_i within clusters and applies $M/(M-1)$, so that the robust variance is the cluster one with one household per cluster and the survey one of a simple random sample. The influence function of $\hat{\beta}$, $\text{IF}_i(\hat{\beta}) = Ps_i$, is what the elasticity standard errors of Section 5.6 are built from, and one aggregator — robust, cluster or survey — serves both.

5.3 The generated regressor

The design contains y , a function of β , and the covariance of Section 3 is that of the last linear step *given* y . The Jacobian of the estimator carries an extra $\partial y/\partial\beta$, and `easi` includes it in closed form. Write the estimator as GMM in β with fixed instruments and residual $e_{im}(\beta) = w_{im} - x_i(y(\beta))'b_m$. The moment vector has blocks $m_j(\beta) = \sum_i w_i \hat{x}_i \sum_m \sigma^{jm} e_{im}(\beta)$, and since $\partial x_i/\partial y = d_i$ and $b_m' d_i = \aleph_{im}$ — the semi-elasticity, already computed for the elasticities —

$$\frac{\partial e_{im}}{\partial b_l'} = -\left[x_i' \mathbf{1}\{m=l\} + \aleph_{im} \frac{\partial y_i}{\partial b_l'} \right].$$

Hence $-\partial m/\partial\beta' = G = (\hat{\Sigma}^{-1} \otimes A) + C$ with row-block j of C equal to $\sum_i w_i v_{ij} \hat{x}_i q_i'$, $v_{ij} = \sum_m \sigma^{jm} \aleph_{im}$ and $q_i = \partial y_i/\partial\beta$; the variance is $P_G \hat{B} P_G'$, and the conditional covariance is the case $C = 0$. Nothing here is numerical: $p'A(z)p$ and $p'Bp$ are *linear* in β , so with $D_i = 1 - p'Bp_i/2$,

$$q_i = \frac{1}{2D_i} (\text{np}_i \otimes g_i),$$

g_i being nonzero only on the price positions of the equation's block. `e(vctype2)` records whether the term is in force.

5.4 Heteroskedasticity

Budget-share equations are heteroskedastic, and the cost of assuming otherwise falls almost entirely on the coefficients in y — the ones that shape the Engel curves — because they are the nonlinear, high-leverage part of the design. Table 3 scores the four combinations of covariance (homoskedastic or sandwich) and Jacobian (conditional on y or with the generated-regressor term) against the bootstrap, on a light specification under simple random sampling: 9 goods, $R = 3$, three demographics, no interactions, 120 coefficients.

Table 3: Bootstrap standard error divided by the analytic one. Above one means the analytic one is too small. *hixdata*, 500 replications.

	terms in y (24)	all others (96)
conventional, conditional on y	1.227 (max 1.69)	1.023 (max 1.19)
conventional, $+\partial y/\partial\beta$	1.227 (max 1.69)	1.023 (max 1.19)
robust , conditional on y	1.007 (max 1.07)	1.002 (max 1.07)
robust, $+\partial y/\partial\beta$	1.007 (max 1.07)	1.003 (max 1.07)

The homoskedastic covariance understates the standard errors of the coefficients in y by 23% on average and by 69% at worst; the sandwich is within 1% on average and 7% at worst. The generated-regressor term moves the standard errors by between -0.5% and $+0.2\%$: y depends on β only through $p'A(z)p/2$ and $p'Bp/2$, and normalised log prices are small. `easi` therefore defaults to `vce(robust)` and keeps the term, because it is the correct Jacobian even where it is small. `vce(conventional)`, which `compat` imposes, returns the homoskedastic covariance that `systemfit`, `reg3` and the R package report.

5.5 Clusters and the survey design

Budget-share data almost always come from a clustered, stratified survey, and the variance has to know it. `easi` reads the design declared by `svyset` — weight, primary sampling unit, strata, finite-population correction — under `vce(svy)`, adopting the survey weight when none is given on the command line and recording the variables in `e(svywvar)`, `e(svyunit)`, `e(svystrata)` and `e(svyfpc)`. The estimator is the stratified ultimate-cluster linearization: the scores s_i are summed within each PSU, the PSU totals u_{hi} are centred on *their own stratum mean*, and stratum h contributes

$$(1 - f_h) \frac{n_h}{n_h - 1} \sum_{i \in h} (u_{hi} - \bar{u}_h)(u_{hi} - \bar{u}_h)'.$$

Two things separate this from the plain cluster sandwich: the centring is per stratum rather than global, and the finite-sample factor is per stratum. Only the first-stage f_h enters, which is exactly what makes the ultimate-cluster approximation legitimate; a value at or below one is read as a sampling rate, anything larger as a population size, following Stata’s convention. Absent an `fpc()`, $f_h = 0$: the PSUs are treated as drawn with replacement, which overstates the variance — safe, but not exact once the sampling fraction is appreciable. A stratum holding a single PSU contributes nothing, and the command says so rather than letting the variance quietly shrink.

Table 4 scores the three variance estimators against the design bootstrap on the Mexican survey.

Table 4: Design-bootstrap standard error divided by the analytic one, 500 replications on the reduced Mexican survey: 2,477 households, 703 PSUs, 28 strata.

	mean	min	max
<code>vce(robust)</code>	1.073	0.907	1.334
<code>vce(cluster psu)</code>	1.016	0.906	1.164
<code>vce(svy)</code>	1.022	0.933	1.181

Ignoring the clusters understates the standard errors by 7% on average and by 33% at worst, and the coefficients it misses most are again those on y . Both design-aware estimators close that gap to within the bootstrap’s own Monte Carlo error; the difference between them, 1.016 and 1.022, is inside it. What `vce(svy)` adds to `vce(cluster)` is that it reads the design rather than requiring the user to know it, and the finite-population correction.

5.6 Every elasticity is a ratio of two estimators

The reported expenditure elasticity (15) is

$$\hat{\eta}_j^x = 1 + \frac{\bar{\aleph}_j}{\bar{w}_j}, \quad \bar{\aleph}_j \equiv \overline{\aleph_j / \Phi},$$

where $\bar{w}_j$ is a mean budget share estimated from the same sample. It is an estimator, and it sits in the denominator. Differentiating with respect to β alone, holding $\bar{w}$ fixed — which is what the Jacobian of Section 4.5 does — keeps one term of three:

$$\text{Var}(\hat{\eta}) = \underbrace{\frac{\text{Var}(\bar{\aleph})}{\bar{w}^2}}_{T_1} + \underbrace{\frac{\bar{\aleph}^2}{\bar{w}^4} \text{Var}(\bar{w})}_{T_2} - \underbrace{\frac{2\bar{\aleph}}{\bar{w}^3} \text{Cov}(\bar{\aleph}, \bar{w})}_{T_3}.$$

The two others are not a small correction, and their sign is not the usual one. $\bar{\aleph}$ and $\bar{w}$ move together, the ratio is steadier than either, and T_1 alone *overstates* the standard error when the

covariance is strong, as it is for goods with small budget shares, and understates it for a good that takes most of the budget.

Every reported elasticity is of the same form: a ratio $\bar{N}/\bar{D}$ of two means over the households, minus one on the diagonal of the price tables (a plain mean $\bar{N}$ for the demographic semi-elasticities), with the weights $\tilde{\omega}$ of its type (Section 4.3). `easi` gives each one its full influence function, in closed form,

$$\text{IF}_h(\hat{\eta}) = \frac{\tilde{\omega}_h}{\sum \tilde{\omega}} \left[\frac{N_h - \bar{N}}{\bar{D}} - \frac{\bar{N}}{\bar{D}^2} (D_h - \bar{D}) \right] + g' \text{IF}_h(\hat{\beta}),$$

where the bracket is the influence of the two sample means with β fixed and the last term the influence of $\hat{\beta}$ through the Jacobian of Section 4.5. For the expenditure elasticity, $N_h = \mathfrak{N}_{hj}/\Phi_h$ and D_h is the observed share. Its variance goes through the same aggregator as everything else, so robust, cluster and survey sampling need no special case. Table 5 scores the result against the bootstrap on the Mexican survey, where the shares are small and the sampling term matters most; Table 6 does the same on *hixdata*.

Table 5: Reported elasticity standard error divided by the bootstrap one, 500 replications, Mexican survey, $R = 3$. Mean shares 0.022, 0.014, 0.964. Under `pweight`, households resampled with their weights; under the survey design, PSUs within strata.

	expenditure			own price		
	good 1	good 2	good 3	good 1	good 2	good 3
<code>pweight, vce(robust)</code>	1.046	0.986	1.006	1.011	0.985	1.015
survey design, <code>vce(svy)</code>	1.013	1.040	1.000	1.002	0.966	0.964

5.7 The elasticities against the bootstrap

Table 6 applies the bootstrap to the elasticities on *hixdata*, that is, to what users publish: the full influence function of Section 5.6, on the robust covariance with the generated-regressor term.

Table 6: Reported elasticity standard error divided by the bootstrap one. *hixdata*, nine goods, the ninth dropped from the system and recovered by adding up; simple random sampling, no interactions; 500 replications.

	median	min	max
expenditure elasticities	0.991	0.955	1.008
own-price elasticities	1.011	0.944	1.042

The dropped good sits inside the same range as the others: its elasticity is a function of β like any other, and the Jacobian of the whole vector carries the negative covariances that adding up creates between equations. The demographic elasticities are checked cell by cell by a different route: without price or expenditure interactions, $\hat{\zeta}_{jt}$ is the coefficient on z_t in equation j , so its standard error must equal $\sqrt{\hat{V}}$ of that coefficient, and $\sqrt{\iota' \hat{V} \iota}$ for the dropped good. It does, to 10^{-12} .

5.8 The influence function by brute force

A bootstrap scores a standard error against another estimate of it, with its own noise, about 3% at 500 replications: it cannot see a term of the variance that weighs 1%. The influence function can be computed without any noise, by brute force. For each household h in turn, the *whole*

procedure is run again with its weight moved to $1 \pm \varepsilon$ ($\varepsilon = 10^{-3}$, `iweights`, not normalised): the mean shares of the instrument, the first pass and the refined instrument, the first stage, $\hat{\Sigma}$, the iteration on y , the final refit, the elasticities. $U_h = \partial \hat{\theta} / \partial w_h$ by central differences is the influence function of every estimate $\hat{\theta}$, with every estimator nested in the procedure moving with the weight, and $\sqrt{n/(n-1) \sum_h U_h^2}$ is the robust standard error in which nothing is neglected. Table 7 divides the reported robust standard errors by it, on two samples of 1,000 households of the design of Section 5.9: 2,000 estimations each.

Table 7: Reported robust standard error divided by the brute-force one, $\sqrt{n/(n-1) \sum_h U_h^2}$, on 1,000 households of the perfect EASI design: median and range over the cells of each block.

		$R = 2$, no interaction	$R = 3$, py zy pz
coefficients		1.0000 (0.9991–1.0003)	1.0000 (0.9982–1.0014)
households	expenditure	0.9998 (0.9997–0.9998)	0.9998 (0.9997–1.0001)
	uncompensated	0.9999 (0.9987–1.0004)	1.0000 (0.9985–1.0003)
	compensated	1.0001 (0.9992–1.0002)	1.0001 (0.9991–1.0002)
	demographic	1.0000 (1.0000–1.0000)	1.0000 (0.9999–1.0001)
market	expenditure	0.9999 (0.9997–1.0001)	0.9993 (0.9992–1.0001)
	uncompensated	0.9999 (0.9984–1.0006)	1.0004 (0.9987–1.0014)
	compensated	1.0001 (0.9992–1.0002)	1.0005 (0.9998–1.0012)
	demographic	1.0000 (1.0000–1.0000)	1.0000 (0.9995–1.0001)

No cell departs by more than 0.16%. The formulas neglect nothing that moves at first order: what is left is the difference between an analytic and a numerical derivative, and the effect of the nested estimators that the formulas treat as fixed — the mean shares that build $\tilde{y}$, the first pass — which is of a higher order.

5.9 A Monte Carlo experiment on data generated by the model

The last check is against the quantity a standard error estimates: the dispersion of the estimator across samples. `dgp_easi.do` draws data from the model itself, in the perfect case — the model holds and every parameter is strongly identified. Three goods; $\ln x \sim N(0, 0.6^2)$ and log prices $\sim N(0, 0.3^2)$ independent across households; a continuous and a binary demographic; $R = 2$; normal errors summing to zero across goods; and y solving the cost identity (4) household by household, with the errors inside the shares, so that $\ln x$ is exogenous and y is not, as in the model. Every coefficient has a t above 10 at $n = 3,000$; the mean shares are about 0.30, 0.25 and 0.45. Seven hundred samples of 3,000 households are estimated under simple random sampling with the default options; the population values are those of the same estimator on 1,000,000 households.

Table 8 reads as follows. A ratio of mean standard error to standard deviation, estimated from 700 samples, has a noise of its own: its 95% band is $\pm 1.96 \sqrt{(k-1)/(4R)}$, k the kurtosis of the estimates, $\pm 5.2\%$ here. None of the 54 cells leaves it. The Wald test of the population value, sample by sample with the analytic standard error of that sample, rejects 5.44% of the time over all cells, against a band of $5\% \pm 1.6\%$ for a single cell; the pooled quantiles of the t statistics are -2.004 and 1.987 . The biases are a tenth of a standard deviation at most. The analytic standard errors are the dispersion of the estimator, for the households and for the market alike.

5.10 The bootstrap as a variance option

`vce(bootstrap[, reps(#) seed(#) strata() psu() svy])` makes the bootstrap of Section 5.1 available to the user. The point estimates are those of the full sample; on each replication the

Table 8: Monte Carlo on the perfect EASI design, 700 samples of 3,000 households: the mean analytic standard error divided by the standard deviation of the estimates across samples (median, min and max over the cells of each block); the largest bias, in standard deviations; the rejection rate of the Wald test at 5% of the population value.

type	block	SE / SD			max bias	Wald
		median	min	max	/ SD	rejection
households	expenditure	1.004	0.963	1.033	0.094	5.19%
	uncompensated	0.990	0.973	1.009	0.110	5.70%
	compensated	0.989	0.970	1.011	0.106	5.67%
	demographic	1.004	0.988	1.010	0.049	4.71%
market	expenditure	0.990	0.961	1.010	0.114	5.81%
	uncompensated	0.993	0.964	1.017	0.114	5.76%
	compensated	0.987	0.968	1.014	0.108	5.57%
	demographic	1.004	0.988	1.010	0.049	4.71%

households (or the primary sampling units of `psu()`, within the strata of `strata()`; `svy` takes both and the weight from `svyset`) are drawn with replacement, keep their weights, and the whole procedure is run again — the prices filled by `pimpute()`, the probits and the completion of Section 6, the iterations. The variance of the coefficients, the standard errors of the elasticities of every type computed and, with `selection`, the joint variance with the probits, are those of the replications; a replication that does not converge is dropped and counted (`e(N_reps_ok)`). The analytic standard errors are the reference when every parameter is well identified, as the tables of this section show; the bootstrap is the safer choice when one is not — typically δ without a variable of the probit only (Section 6.7).

6 Households that do not buy

On a household survey the finer the goods, the more households buy none of some of them in the recording period: nine percent of the households of the Mexican survey buy no corn, fifteen percent no wheat. Two things then go wrong at once. The price of a good is usually its unit value, expenditure over quantity, and a household that buys none has no price; and its zero share is not the share of the model, whose errors are normal and whose shares are never exactly zero. `easi` handles the first with `pimpute()` and the second with `selection`, the two-step estimator of Shonkwiler and Yen (1999) applied to EASI as by Zhen et al. (2014); the options, their names and their rules are those of the author’s `equaids` and `duvm`.

6.1 Missing prices

Without `pimpute()`, a household with a missing price leaves the sample — with unit values, exactly the households that do not buy, which makes any correction for them impossible. `pimpute(varlist)` replaces each missing log price by the weighted mean of the log prices of the households of the same group of the first variable (the primary sampling unit, typically); where nobody in the group has a price, the group of the next variable is used (urban or rural area, say), and so on. Households still without a price leave the sample, and the means are computed again until the sample no longer changes, so that the donors are the households of the final sample. The weights are those of the estimation. A note reports, good by good, how many prices were filled at each level. The analytic standard errors include the imputation. A filled price is the weighted mean m_{gj} of the log prices of the donors of its group, so the weight of a donor d moves it by $\omega_d(\ell_{dj} - m_{gj})/W_{gj}$ (W_{gj} the weight of the donors); the influence function of the donor gains

that change times the derivative, with respect to the filled prices of the group, of the moments of the system, of the scores of the probits and of the sample terms of the elasticities. The term sums to zero within each group, so it cancels under `vce(cluster)` or `vce(svy)` at the level of the first grouping variable. Against a brute force (the whole procedure re-run with the weight of each household moved), the analytic standard errors are within 0.2% with the term; without it, under the correction for the non-buyers of Section 6.2, where the filled prices are those of the non-buyers, the standard errors of some elasticities were up to 9% too large. The bootstrap of Section 5.10 fills the prices again on every replication. `predict` and `estat engel` fill them in the same way.

6.2 The censored system

Let $C \subseteq \{1, \dots, J-1\}$ be the goods corrected: by default every good with zero shares in the sample, the last excepted, or those of `selgoods()`. For $i \in C$ the latent share is the share of the model, $w_i^* = f_i(y, \ell, z) + u_i$ with f_i the right-hand side of (5); the household buys when $s_i' \alpha_i + v_i > 0$, with

$$s_i = (1, \ell_1 - \ln x, \dots, \ell_J - \ln x, z, q_i),$$

q_i the variables of the probit only (`selvars()`, the exclusion restrictions) and $v_i \sim N(0, 1)$; the observed share is $w_i = d_i w_i^*$. With (u_i, v_i) jointly normal and $\delta_i \equiv \text{Cov}(u_i, v_i)$,

$$\mathbb{E}[w_i | s_i, y] = \Phi(s_i' \alpha_i) f_i + \delta_i \phi(s_i' \alpha_i), \quad (19)$$

and $\delta_i = 0$ is the absence of selection. (In this section Φ and ϕ are the standard normal distribution and density; the derivative of the cost function written Φ in (11) is written D .) The last good closes the system, $\mathbb{E}[w_J] = 1 - \sum_{i < J} \mathbb{E}[w_i]$: it must be bought by every household (the rest of the budget, typically), and `easi` refuses a last good with zeros. The probit is written in prices relative to expenditure, so it is homogeneous of degree zero, and so are the expected shares. The two-step estimator runs one weighted probit per corrected good, then the system (19) on all the households, buyers or not; it is consistent, not efficient, since the probits ignore the correlation between the goods.

6.3 The implicit utility of a household that does not buy

y is computed from the shares (4). For a buyer the observed share is the latent one; for a non-buyer of good i the observed zero is not, and the Stone index built on it is off by $\ell_i w_i^*$. `easi` replaces the zero by the expectation of the latent share given no purchase, from the same normal model,

$$\tilde{w}_i = \mathbb{E}[w_i^* | d_i = 0, s_i] = f_i - \delta_i \frac{\phi(s_i' \alpha_i)}{1 - \Phi(s_i' \alpha_i)}, \quad (20)$$

the last good absorbing the difference so that the completed shares still sum to one. With $\text{np}_i = \ell_i - \ell_J$,

$$y = \frac{y^{\text{stone}} - \sum_{i \in C} (1 - d_i) \text{np}_i \tilde{w}_i + \frac{1}{2} p' A(z) p}{1 - \frac{1}{2} p' B p},$$

a fixed point in y , since f_i depends on it, solved at each step of the estimator as (7) is. The error in y becomes $\text{np}_i(w_i^* - \tilde{w}_i)$, of mean zero. y now depends on δ and on the probit coefficients as well as on β , and the variance accounts for it (Section 6.5). The two-step method has been applied to a censored EASI system before (Zhen et al., 2014); the completion of y is an extension. One approximation remains, and it is that of any two-step censored system with an endogenous index: the moment of (19), conditional on the instruments, holds up to a term $\text{Cov}(d_i, f_i(y))$, since y depends on the purchase through the completion. Its order is $b_1^i \text{np}_i \delta_i \phi_i$: on the design of Section 6.8, with $|b_1^i| \leq 0.06$, np_i of standard deviation 0.4, $\delta_i = 0.04$ and $\phi_i \leq 0.4$, a few 10^{-4} on shares of about 0.3.

6.4 The estimator

The equations no longer share their regressors: the regressors of a corrected good are those of EASI times Φ_i , and ϕ_i ; its instruments, those of EASI times Φ_i , and ϕ_i ; an uncorrected good keeps $x(y)$ and z . `easi` solves the restricted GMM problem with moments $g_i = \sum_h w_h Z_{ih} e_{ih}$, $e_{ih} = w_{ih} - X'_{ih}\theta_i$, stacked and weighted by Ω^{-1} , $\Omega_{ij} = \hat{\sigma}_{ij} Z'_i W Z_j$, under the symmetry restrictions of Section 3 (bordered as there): with $M = \text{diag}(X'_i W Z_i)$ and $H = M\Omega^{-1}M'$ the step is $H\theta = M\Omega^{-1}\text{vec}(Z'_i W w_i)$. When X and Z are the same in every equation, $H = \hat{\Sigma}^{-1} \otimes A$ and this is the 3SLS of Section 3; run without selection the engine reproduces it, coefficients, $\hat{\Sigma}$, variances and elasticities, to the precision of the central differences. (The “second stage on the fitted $\hat{X}$ ” form of 3SLS does not converge here: the residual of the projection on z is not orthogonal to $\Phi_i z$.) The loops on y and on $\hat{\Sigma}$ are those of Section 3. Symmetry is imposed on the latent shares f , not on the expected shares, which have no symmetry to impose.

6.5 Variance

Write $\hat{\theta} - \theta = PL(\sum_h g_h + D_\alpha(\hat{\alpha} - \alpha))$ with $L = M\Omega^{-1}$, P the restricted inverse of LD_θ , $D_\theta = -\partial \sum g / \partial \theta'$ taken with y recomputed from θ (the generated regressor), and $D_\alpha = \partial \sum g / \partial \alpha'$ the response of the moments to the probit coefficients, through Φ , ϕ and y . The influence function of the system is therefore

$$\text{IF}_h(\hat{\theta}) = PL(g_h + D_\alpha \text{IF}_h(\hat{\alpha})),$$

the two Jacobians taken by central differences, and it goes through the aggregator of Section 5 (robust, cluster, survey design), stacked with the influence functions of the probits so that `e(V_sel)` holds the joint variance of the coefficients and of the probits.

The influence function of a probit uses the observed Hessian. With $q = 2d - 1$ and $\lambda = q\phi(qs'\alpha)/\Phi(qs'\alpha)$,

$$\text{IF}_h(\hat{\alpha}) = \left[\sum_h w_h \lambda_h (\lambda_h + s'_h \hat{\alpha}) s_h s'_h \right]^{-1} w_h \lambda_h s_h,$$

the derivative of the score the estimator actually solves. The expected (Fisher) information that the scoring iterations use equals it only in expectation under a correct probit; on survey data the two differ at first order, and the first implementation, which used the expected one, was off by up to 4.6% of the standard errors of the system against the brute force of Section 6.8.

6.6 The elasticities of the expected demand

The elasticities are those of the expected demand of all the households, buyers and non-buyers: those of (19). With $\text{xb}_i = s'_i \alpha_i$ and the latent derivatives of Section 4,

$$\begin{aligned} \frac{\partial \mathbb{E}[w_i]}{\partial \ln x} &= \Phi_i \frac{\partial f_i}{\partial \ln x} - \phi_i (f_i - \delta_i \text{xb}_i) \sum_k \alpha_{ik}, \\ \frac{\partial \mathbb{E}[w_i]}{\partial \ell_k} &= \Phi_i \frac{\partial f_i}{\partial \ell_k} + \phi_i (f_i - \delta_i \text{xb}_i) \alpha_{ik}, \quad \frac{\partial \mathbb{E}[w_i]}{\partial z_t} = \Phi_i \frac{\partial f_i}{\partial z_t} + \phi_i (f_i - \delta_i \text{xb}_i) \alpha_{izt}, \end{aligned}$$

α_{ik} the coefficient of $\ell_k - \ln x$ in the probit, and y responding through the completed shares, $\partial y / \partial \ell_k = -\tilde{w}_k / D$ and $\partial y / \partial \ln x = 1 / D$, D the derivative (11) of the cost function evaluated at the completed shares. The three types of Section 4.3 are the same means, with the expected shares in the denominators, so that Engel and Cournot aggregation hold exactly again; the compensated elasticities follow from the Slutsky equation household by household, $\eta_{hjk}^c = \eta_{hjk} + \mathbb{E}[w_{hk}] \eta_{hj}^x$. Their standard errors carry the sampling term of Section 5.6, the Jacobian in θ and the Jacobian in α times the influence functions of the probits.

These formulas are checked against finite differences of the expected demand, household by household: $\mathbb{E}[w]$ recomputed at $\ln x \pm h$ and $\ell_k \pm h$ with y re-solved from the completed cost

identity, the latent residuals held fixed, and averaged with the weights of each type. The reported elasticities agree to 2×10^{-11} — expenditure, uncompensated, compensated and demographic, households and market — on the Mexican survey with sampling weights and a variable of the probit only.

6.7 Identification of δ

δ_i is identified by the variables of the probit only or, without them, by the nonlinearity of the normal distribution alone: ϕ_i is then close to a linear combination of the columns $\Phi_i x$. **easi** measures how close with the variance inflation of δ_i ,

$$\text{VIF}(\delta_i) = [(X_i' W X_i)^{-1}]_{\delta\delta} \phi_i' W \phi_i = \frac{1}{1 - R_i^2},$$

R_i^2 the uncentred R^2 of ϕ_i on the other regressors of its equation. A table under the header reports, for each corrected good, the percentage of buyers, the pseudo- R^2 of its probit, the households predicted with probability 0 or 1 (separation), $\text{VIF}(\delta_i)$ and δ_i with its z ; above 10, a warning says that δ_i rests on little more than the curvature of the probit, and suggests leaving the good uncorrected (**selgoods()**), adding a variable of the probit only (**selvars()**), or the bootstrap. Under the table, the size of the completion of y : the percentage of the households it touches and the mean absolute change of their y , in units and in standard deviations of y . **easidiag** reports the same table before any estimation, at the starting point $y = y^{\text{stone}}$ (Section 7).

6.8 Validation

The estimator is checked against closed forms first. With two goods, power one and no interaction, the system is one exactly identified equation and **easi** with **selection** is Stata's **ivregress 2sls** of w_1 on $(\Phi, \Phi y, \Phi z, \Phi np, \phi)$ with instruments $(\Phi, \Phi z, \Phi np, \Phi \tilde{y}, \phi)$, Φ and ϕ from Stata's **probit** and y the completed implicit utility in closed form: the coefficients and δ agree to 2×10^{-11} , **predict** returns $\mathbb{E}[w]$, the latent shares, y and the residuals to 10^{-11} , and the size of the completion to 2×10^{-12} ; the probits reproduce Stata's **probit** to 4×10^{-10} . Without selection the general engine gives the coefficients, $\hat{\Sigma}$ and the elasticities of the 3SLS engine to 2×10^{-11} , and its variances, whose Jacobians it takes by central differences, to 4×10^{-8} , on five specifications.

The variance is then scored three ways (Table 9). The brute force of Section 5.8 on 800 households of the Mexican survey, without variable of the probit only — the weakly identified case, the most demanding for a linearization; a bootstrap of the whole procedure (probits, completion, system), 500 draws on one sample of 10,000 households of the perfect censored design; and a Monte Carlo of 700 samples of that design. The design extends that of Section 5.9: goods 1 and 2 bought with probabilities of about 72% and 73% by probits with an exclusion variable each, $\rho = 0.8$ between the errors of the share and of the probit, $\delta = 0.04$, and the latent intercepts raised by 0.03 so that a buyer's latent share is almost never negative (0.8 households per sample of 10,000 on average, which **easi** reads as non-buyers).

Table 9 reads as the tables of Section 5. The brute force, on real data and in the weakly identified case, finds nothing neglected: every standard error within 0.5%, with the estimation of the probits moving with the weight like the rest (the standard errors of the probit coefficients themselves are not scored this way). The bootstrap, whose noise is $\pm 6\%$ for one cell at 500 draws, puts 2 of the 84 cells outside that band, and its median ratio over all cells is 1.006. The Monte Carlo puts 2 of the 84 cells outside its band of $\pm 5.3\%$, with a median ratio of 0.989 and a rejection rate of 5.48% over all cells. δ is the one parameter with a visible finite-sample bias, a third of its standard deviation against the value of the same estimator on 500,000 households (whose δ are 0.039 and 0.040, for a true 0.04), which raises the rejection rate of its Wald test to

Table 9: The variance under selection. Brute force: reported robust standard error divided by $\sqrt{n/(n-1)\sum_h U_h^2}$, 800 households of the Mexican survey, no variable of the probit only. Bootstrap: analytic standard error divided by the bootstrap one, 500 draws on one sample of 10,000 households of the censored perfect design. Monte Carlo: mean analytic standard error divided by the standard deviation of the estimates, 698 samples of 10,000 households of the same design (two of 700 did not converge), and the rejection rate of the Wald test at 5% of the population value. Median and range over the cells of each block.

		brute force	bootstrap	Monte Carlo	
				SE / SD	Wald
latent coefficients		0.999 (0.996–1.001)	0.970 (0.939–1.052)	0.996 (0.961–1.044)	5.2%
δ		0.998 (0.998–0.999)	0.985 (0.976–0.993)	0.986 (0.969–1.003)	6.3%
probit coefficients		—	1.006 (0.940–1.068)	0.993 (0.933–1.031)	5.6%
households	expenditure	1.001 (1.001–1.001)	1.051 (1.029–1.051)	1.012 (0.969–1.049)	4.8%
	uncompensated	1.001 (0.999–1.003)	1.006 (0.971–1.040)	0.988 (0.949–1.025)	5.9%
	compensated	1.001 (0.999–1.003)	0.996 (0.969–1.023)	0.983 (0.945–1.031)	5.8%
	demographic	1.000 (0.999–1.000)	1.021 (0.948–1.046)	0.987 (0.979–1.050)	5.0%
market	expenditure	1.000 (1.000–1.005)	1.043 (1.033–1.058)	1.018 (1.002–1.048)	4.8%
	uncompensated	1.000 (0.999–1.003)	1.006 (0.968–1.045)	0.984 (0.956–1.030)	5.8%
	compensated	1.000 (0.999–1.003)	0.999 (0.971–1.027)	0.984 (0.946–1.037)	5.7%
	demographic	1.000 (0.999–1.001)	1.020 (0.954–1.047)	0.988 (0.981–1.051)	4.9%

6.3%; a two-step estimator that plugs estimated probits into a nonlinear system is not expected to do better at $n = 10,000$.

6.9 The Mexican cereals

The reduced Mexican survey has no strong exclusion restriction, and that is worth showing. Table 10 runs the correction for corn and wheat under the survey design, first through `easidiag`. The probit of wheat barely separates buyers from non-buyers (pseudo- R^2 0.017), and without a variable of the probit only ϕ is almost a combination of the Φx columns: $\text{VIF}(\delta)$ is 1,982. The age and the sex of the head, in the probits only, bring it to 104 for wheat and 24 for corn — still above 10, and `easi` warns. Neither δ differs from zero, and the correction moves the elasticities of the households by less than their standard errors; it moves those of the market more (corn, from 0.068 to 0.098). The completion touches 23.5% of the households and moves their y by 0.015 on average, 0.019 standard deviations.

A bootstrap of the whole procedure, 500 replications of the PSUs within the strata, shows what weak identification does to the linearization. The survey standard errors of the probit coefficients and of the elasticities of the households are within the noise of the bootstrap (medians 0.998 and 0.98 to 1.04 of it); those of δ are 10% larger than the bootstrap ones on the median and up to 29%, and a few latent coefficients depart by as much, in both directions. This is the case for `vce(bootstrap)`: 500 replications took three minutes.

7 Diagnosing a specification before estimating it: `easidiag`

Two things make an EASI system fail to converge, or converge to something meaningless: a regressor with almost no variance, and a near-linear dependency among regressors. Both are invisible in the estimator’s own output — the iteration log shows a criterion that will not fall, and nothing else. A diagnostic that needed the model to fit would be silent exactly when the model does not fit, so `easidiag` computes everything from the data and from the design matrix evaluated at the starting point $y = y^{\text{stone}}$, a function of the data alone. It takes the syntax of

Table 10: The households that do not buy, reduced Mexican survey (2,477 households, $R = 3$, `vce(svy)`; 9% buy no corn, 15% no wheat). Top: the probits and $\text{VIF}(\delta)$ from `easidiag`, at $y = y^{\text{stone}}$, and δ from `easi` with the age and the sex of the head in the probits only. Bottom: expenditure and own-price elasticities without and with the correction (standard errors in parentheses).

	buyers	pseudo- R^2	$\text{VIF}(\delta)$, no exclusion	$\text{VIF}(\delta)$, age, sex	$\delta(z)$
corn	91.3%	0.132	33.3	23.7	0.0304 (1.39)
wheat	84.7%	0.017	1,982.3	104.0	-0.0137 (-0.89)

		expenditure		own price	
		none	corrected	none	corrected
households	corn	0.1214	0.1136 (0.039)	-0.9647	-0.9611 (0.042)
	wheat	0.4751	0.4794 (0.037)	-0.7106	-0.7153 (0.063)
	other	1.0276	1.0277 (0.001)	-1.0224	-1.0227 (0.002)
market	corn	0.0679	0.0984	-0.9512	-1.0102
	wheat	0.4181	0.4277	-0.5881	-0.6227
	other	1.0162	1.0164	-1.0116	-1.0128

`easi`, weights, `hhszize()`, `pimpute()` and the selection options included, accepts and ignores the estimator-only options (and says which), and reports eight sections, nine with `selection`.

Why condition indexes, not rank. An exact dependency is the benign case: the generalised inverse pins one column at zero and the identified quantities come out right. It is the *near*-exact dependency that destroys the estimate, and a rank test does not see it. Table 11 perturbs the sum of an exhaustive set of household-composition dummies on the reduced Mexican survey.

Table 11: An exhaustive set of dummies, with the constant, perturbed away from exact collinearity. Reduced Mexican survey, three goods, $R = 3$.

perturbation of the dummy sum	max coefficient	max std. error
0 (exact)	0.0229	0.0014
10^{-12}	0.0229	0.0014
10^{-9}	622.07	58 715

A `float` carries a relative error of 6×10^{-8} , sixty times what it takes. `easidiag` therefore measures the condition indexes of [Belsley et al. \(1980\)](#), which are continuous, and reports the variance-decomposition proportions, which name the columns involved; and it flags any non-integer regressor stored as `float`.

1. Data admissibility. Shares summing to one, shares outside $[0, 1]$, observations lost to missing values.

2. Screening of the design columns. The standard deviation and the variance inflation factor of every column of the design, including the constructed ones — the powers of y , the normalised log prices and all the interactions. Then the condition indexes: a component whose index exceeds 30 and on which two or more columns carry more than half their variance is a

near-dependency, and the proportions name the columns. A group that includes the constant is the classic exhaustive set of category dummies, and the report says so:

```
near-dependency, condition index effectively infinite :
_cons nocup0 nocup1 nocup2 nocup3 nocup4
(this group sums to a constant: an exhaustive set of dummies)
```

3. Categorical and discrete variables. For each demographic: the number of levels, the storage type, and for a binary variable the size of the smaller cell. A cell below 2% is flagged, because under `pz` such a variable contributes $J - 1$ price interactions that are non-zero for a handful of households; merging categories is usually the answer. A variable with a few integer levels is flagged as entering *linearly*, which imposes that a move from level 1 to 2 has the same effect as from 10 to 11.

4. Normalisation and conditioning. The mean, the standard deviation and the ratio $|\text{mean}|/\text{sd}$ of log expenditure and of every log price, then the condition number of $X'X$ as given and as it would be if the variables were centred. EASI is written for prices and expenditure measured around a base period. Fed raw levels, the columns $1, y, y^2, y^3$ correlate above 0.999 and the normal matrix goes singular: on the Mexican survey, $\text{cond}(X'X)$ was 4.5×10^{10} in levels and 4.5×10^1 centred.

5. Structural sources of a flat objective. The conditioning of the Gram matrix of $1, y, \dots, y^R$, reported for every power up to `power()`: centred, each extra power costs a factor of about four; uncentred, about a hundred. The number of *distinct price vectors*: when prices are regional or temporal indexes, every household in a cell shares one price vector and the price parameters are identified off the number of cells, not the sample size — on *hixdata*, 48 distinct price vectors for 4,847 households, against 36 free parameters in the symmetric block A alone. The rank of the price variation, the first-stage R^2 for y , and the conditioning of the covariance of the observed shares, which stands in for the residual covariance before there are any residuals.

6. Scale and variability. Mean and standard deviation of each budget share, and the standard deviation of each normalised log price. A mean share below 0.01 is flagged: every semi-elasticity of that good is divided by it.

7. Convergence of the fixed point. $D = 1 - p'Bp/2$ is the denominator of y ; as it approaches zero the map stops contracting, and a negative D means the cost function has left its regular region. The contraction factor is measured by taking two steps of (7) from the 2SLS start, $\rho = |y_2 - y_1|/|y_1 - y_0|$; if $\rho < 1$ the number of iterations needed is $\log(\text{tol}/|y_1 - y_0|)/\log \rho$. This is the one section that touches the model, and it is skipped when the screening has already found something that stops the design being solved. On *hixdata* the prediction is exact: 4 predicted, 4 observed.

8. Identification. Ranks of $Z'WZ$ and of $\hat{X}'W\hat{X}$, and the count of instruments against coefficients per equation.

9. Selection of the buyers. With `selection`: for each corrected good, the probit that `easi` would estimate — same sample, same weights, the prices filled by `pimpute()` — with the percentage of buyers, its pseudo- R^2 , the households predicted with probability 0 or 1, the number of variables of the probit only, and $\text{VIF}(\delta)$ of Section 6.7 at $y = y^{\text{stone}}$; above 10 the good is flagged. On the Mexican survey without a variable of the probit only it reads 33 for corn and

1,982 for wheat (Section 6.9): the correction would rest on the curvature of the probit, and the report says so before anything is estimated.

7.1 Using the diagnostic: from a system that will not converge to one that does

The practitioner’s experience of EASI is that it converges beautifully on the textbook data and then, on one’s own survey, either stops at the iteration limit with a criterion that will not fall, or converges to elasticities with standard errors in the thousands. The estimator’s own output gives no clue: the log shows the criterion, the coefficient table is full of numbers, and nothing says which variable is responsible. The remedy is almost always a change of specification rather than of algorithm, and the point of `easidiag` is to say *which* change, before the first estimation is launched. Table 12 maps what the report shows to what to do.

Table 12: Reading an `easidiag` report.

what the report says	what it means	what to do
section 4: not centred , <code>cond(X'X)</code> above 10^6 as given but small if centred	prices or expenditure are in raw levels; $1, y, y^2, y^3$ are near-collinear because y sits far from zero	subtract the sample mean from log expenditure and from every log price (or normalise to a base period); no change of model
section 5: polynomial basis T00 HIGH at the chosen power, even centred	the Engel-curve polynomial is over-parameterised for the range of y	lower <code>power()</code> ; each extra power costs about a factor of four in conditioning when centred
section 2: near-dependency, <i>condition index effectively infinite</i> , group contains <code>_cons</code> and a set of dummies	an exhaustive set of category dummies sums to the constant	drop one category; the estimator would silently pin one coefficient to zero, but any rounding in a <code>float</code> variable turns that into standard errors of 10^5
section 2: near-dependency with a finite index (30–100), a group of interactions	mild redundancy among interactions, usually <code>price×expenditure</code> or <code>time×y</code>	a warning, not a stop: the model fits; if the standard errors of that block are large, drop the interaction that carries no economics
section 3: rare , merge it (a binary with a cell below 2%)	under <code>pz</code> that variable adds $J - 1$ price interactions identified off a few dozen households	merge the category with its neighbour, or leave it out of <code>interpz()</code>
section 3: entered linearly (cardinal)	a variable with a few integer levels (region, month) enters as a number	replace by dummies, dropping one, unless the effect really is proportional to the code
section 5: distinct price vectors FEW	prices are regional or temporal indexes; the price block is identified off the number of cells, not the sample size	keep the price block lean: no <code>pz</code> , or <code>interpz()</code> on one or two demographics; aggregate goods
section 6: mean share SMALL	every elasticity of that good is a ratio with a denominator near zero	aggregate the good with a neighbour; if it must stay, expect wide intervals and read Section 5.6
section 7: $\Phi \leq 0$ for some households, or contraction factor near one	the cost function leaves its regular region for those households; the fixed point crawls	lower <code>power()</code> or drop <code>py</code> ; check the households with extreme relative prices
section 8: rank DEFICIENT	a regressor is an exact function of the others	the group is named in section 2; remove one column of it

Four situations show the report at work; the first three are run on the reduced Mexican survey shipped with the module (2,477 households, three cereal goods, `mex_bench.dta`), the last

on *hixdata*. All four are reproduced by `replication/Table11.do`.

Raw levels. Log prices and log expenditure as they come from the survey, means 2.8, 3.9 and 8.2. The report flags the polynomial and names the cause:

```
near-dependency, condition index 4099 :
  _cons y1 y2 y3
lnexpenditure 8.2438 0.8124 10.15 not centred
cond(X'X) as given 1.680e+07 ILL-COND
cond(X'X) if centred 2.296e+05
Verdict: 4 point(s) to look at.
2. 3 variable(s) are not centred; subtract their means
4. the y polynomial is ill-conditioned at power(3);
   centre y or lower the power
```

After subtracting the means the same specification reads `cond(X'X) = 5.3e+01`, Verdict: nothing to flag. The specification should estimate cleanly in about 3 iterations — and it does, in three.

An exhaustive set of dummies. Five household-composition dummies that sum to one, all entered:

```
near-dependency, condition index effectively infinite :
  _cons nocup0 nocup1 nocup2 nocup3 nocup4
  (this group sums to a constant: an exhaustive set of dummies)
rank of Xhat'W Xhat 10 / 11 DEFICIENT
2. drop one category from the exhaustive dummy set
```

`easi` itself converges on this specification without a word — the generalised inverse pins one coefficient at zero — which is why the diagnostic exists. Perturb the dummies by 10^{-9} , as storing them as `float` would, and the same report appears while the estimator now returns standard errors of 5.9×10^4 (Table 11).

A rare category under pz. A binary demographic with 1.2% of ones, crossed with prices:

```
rare 2 byte 0.0125 rare, merge it
Verdict: 1 point(s) to look at.
1. 1 binary demographic(s) have a cell below 2%; merge categories
```

The design is not ill-conditioned and the fixed point contracts, so the estimation will run; the report says that the $J - 1$ price interactions of that variable rest on some thirty households.

The reference specification. *hixdata*, nine goods, five demographics, $R = 5$, all interactions, 576 parameters — the model of the vignette, which every implementation is judged on:

```
near-dependency, condition index 52 :
  time#y np_srent#y np_sfurn#y np_srecre#y
time 12 long --- entered linearly (cardinal)
distinct price vectors 48 / 4847 FEW
contraction factor |y2-y1|/|y1-y0| 0.0315 ok
iterations predicted to reach 1.0e-06 4
Verdict: 3 point(s) to look at.
```

Nothing here stops the estimation, and it converges in three to four iterations as predicted. But the report says three things about the reference model that its published tables do not: `time` is a twelve-level calendar code entered as a number; the 36 free parameters of the symmetric price block are identified off 48 distinct price vectors, not 4,847 households; and the `price`×`expenditure` interactions carry a mild redundancy. A practitioner adapting this specification to a survey with

regional price indexes — typically a dozen cells — would want to know all three before choosing `pz`.

`easidiag` stores the counts and the key scalars in `r()`: `r(nprob)`, `r(cond_raw)`, `r(cond_ctr)`, `r(ratio)`, `r(iter)`, `r(phimin)`, `r(wmin)`, the matrices `r(var)` and `r(good)`, and with `selection` `r(selgoods)` and `r(sel_diag)`. It is also reached from the dialog box, `db easi`, under *Action: 2 – Diagnostic*.

8 The easi command

8.1 Syntax

```
easi sharevars [if] [in] [weight],
    {prices(varlist) | lnprices(varlist)}
    {expenditure(varname) | lnexpenditure(varname)}
    demographics(varlist) [options]
```

The share variables must sum to one; the last one is dropped from the system and recovered by adding up. `prices()` takes prices in levels and logs them; `lnprices()` takes logarithms; likewise for expenditure. At least one demographic is required.

Model. `power(#)` sets R (default 5). `py`, `pz` and `zy` add the price×expenditure, price×demographic and demographic×expenditure interactions; `interpz(varlist)` restricts the second to named demographics (default: all). The legacy spellings `inpy(1)`, `inpz(1)`, `inzy(1)` are accepted.

Missing prices and non-buyers. `pimpute(varlist)`, `selection`, `selgoods(namelist)` and `selvars([good:] varlist [; ...])` (Section 6); `selgoods()` and `selvars()` imply `selection`.

SE/Robust. `vce(robust)` (the default), `vce(cluster clustvar)`, `vce(svy)`, `vce(conventional)` and `vce(bootstrap [, reps() seed() strata() psu() svy])` (Sections 5 and 5.10).

Elasticities. `elasticities(households | individuals | market)` and `hysize(varname)` (Section 4.3).

Reporting. `snames(namelist)` labels the goods; `dec(#)` sets the decimals (default 4); `dislas(0)` hides the dropped good; `dregres(1)` also shows the coefficient table; `compensated`, `demoelast` and `checks` add the compensated price elasticities, the demographic elasticities and the aggregation identities, and `detail` adds all three; `noelastse` skips the elasticity standard errors and their tables; `stars` marks the elasticities with significance stars; `saveres(filename)` writes every table to a file whose extension sets the format (`.docx`, `.tex`, `.xlsx`, `.csv`, `.md`); `notable` shows the header only; `nolog` suppresses the iteration log. The reporting options, and `elasticities()`, can be given on replay, `easi`, `compensated saveres(tables.docx)`.

Advanced. `tolerance(#)` (default 10^{-6}) and `iterate(#)` (default 100); `compat` (Appendix A).

`aweights`, `pweights` and `iweights` are allowed; `fweights` are refused (Section 3.5). The lists are checked before anything is estimated: a variable may appear once in a list and in one role only, and every good named in an option must be one of the shares.

8.2 Output

The command prints the iteration log, a header with the sample, the design, the type of elasticities, the variance estimator and the execution time (`e(time)`, measured with Stata timer 100); with `selection`, the table of the probits of Section 6.7; then the expenditure elasticities, the uncompensated price elasticities as a $J \times J$ block, and, on request, the compensated block, the demographic elasticities and the aggregation identities. Each elasticity table is followed by the table of its standard errors. The tables are drawn by a private copy of the author's `tabstars` engine, the same from Stata 14.2 to 19: the $J \times J$ blocks are printed in one piece, whatever `linesize`, with the good names intact, scrolling sideways in the Results window, and the same tables are written to Word, L^AT_EX, Excel, CSV or Markdown by `saveres()`, with or without `stars`.

8.3 Stored results

`easi` is `e`-class. Scalars: `e(N)`, `e(N_eff)`, `e(ngoods)`, `e(neq)`, `e(nsoc)`, `e(power)`, `e(k_eq)`, `e(py)`, `e(pz)`, `e(zy)`, `e(iter)`, `e(crit)`, `e(converged)`, `e(time)`, `e(N_psu)`, `e(N_strata)`, `e(chk_engel)`, `e(chk_cournot)` and their `_mkt` twins; `e(N_reps)`, `e(N_reps_ok)` with `vce(bootstrap)`; `e(sel_ycomp_pct)`, `e(sel_ycomp_mean)`, `e(sel_ycomp_sd)` with `selection`. Macros: `e(cmd)`, `e(shares)`, `e(prices)`, `e(demographics)`, `e(interpz)`, `e(vce)`, `e(clustvar)`, `e(svyunit)`, `e(svystrata)`, `e(svywvar)`, `e(svyfpc)`, `e(vcetype2)`, `e(wtype)`, `e(wexp)`, `e(mode)`, `e(report)`, `e(predict)`, `e(elasticities)`, `e(hhsize)`, `e(pimpute)`, `e(boot_design)`, `e(boot_seed)`, and with `selection` `e(selection)`, `e(selgoods)`, `e(selvars)`, `e(sel_z_good)`. Matrices: `e(b)`, `e(V)`, `e(elast_exp)` and `e(elast_exp_se)`, `e(elast_price_nc)` and `e(elast_price_nc_se)`, `e(elast_price_c)` and `e(elast_price_c_se)` (the latter filled only with compensated), `e(elast_demo)` and `e(elast_demo_se)`, `e(semi_exp)`, `e(semi_price)`, `e(slutsky)`, `e(Sigma)`; the market elasticities and their standard errors under the same names followed by `_mkt`; with `selection`, `e(sel_alpha)` (the probit coefficients), `e(sel_diag)` (buyers, pseudo- R^2 , separation, $VIF(\delta)$) and `e(V_sel)` (the joint variance of the coefficients and of the probits); and, kept unchanged for existing do-files, `e(elast_income)`, `e(elast_price)` (the transpose), `e(semi_income)` and `e(compensated_q)` ($\hat{\Gamma}$, the compensated elasticities plus the identity). `test`, `lincom`, `nlcom` and `estimates` work as usual.

8.4 Postestimation

```
predict stub*, shares [latent] | residuals
predict newvar, y
```

Everything is recomputed from `e(b)`, including the implicit utility index (which has the closed form (4), so no iteration is needed), and predictions are available outside the estimation sample. The fitted shares sum to one exactly, the last being obtained by adding up. After `selection`, `shares` are the expected shares (19), `latent` gives the latent shares f instead, `residuals` are $w - \mathbb{E}[w]$ and `y` is the completed implicit utility of Section 6.3, the probits read from `e(sel_alpha)`.

`estat engel` traces the fitted budget shares against total expenditure. By default (`atmeans`) the demographics and the prices are held at their weighted means and the fixed point in (w, y) is solved at each grid point, so the curve is the exact function and `n()` is a resolution rather than a smoothing parameter. `asobserved` instead leaves the covariates as observed, which is a scatter and is smoothed by a local linear regression; there `bwidth()` applies, with the rule of thumb for local polynomial *regression* as the default — Silverman's rule is a density rule and is not the right one for a smoother — and one bandwidth serves every panel so that they are comparable. The confidence band is the delta-method standard error $x'\hat{V}x$ of the fitted share, with the full sum of covariance blocks for the dropped good. A polynomial of order `power()`

oscillates near the ends of the range of y , so `trim()` drops one percent from each tail of the grid by default, and the figure says so. `data()`, `saving()`, `level()`, `noci` and `nodraw` are available. After `selection` the curves are those of the expected shares, the probits at the means of prices, demographics and variables of the probit only as well, and the band includes the estimation of the probits (`e(V_sel)`).

8.5 Dialog box

`db easi` opens a dialog with an *Action* selector: *1 – Estimation* builds the `easi` command, *2 – Diagnostic* builds the `easidiag` command from the same fields, greying out the estimator-only controls. The tabs cover the model, the interactions, the variance estimator (the bootstrap with its replications, seed and survey design), the type of elasticities, the households that do not buy (missing prices, the goods corrected, the variables of the probits, for every good or good by good) and the reporting options, `compat` included; the dialog accepts prices and expenditure in levels. Nine examples of the help file run from a link, in the command window, as a do-file, and — all but the one on `predict` and `estat engel` — in the dialog box filled in, with prices and expenditure in levels built from the logs of the example data (`easi_examples # [, db | do]`); the data in memory are kept.

8.6 Example

```
. use hixdata, clear
. easi sfoodh sfoodr srent soper sfurn scloth stranop srecre spers,
    lnprices(pfoodh pfoodr prent poper pfurn pcloth ptranop precr ppers)
    lnexpenditure(log_y) demographics(age hsex carown time tran)
    power(5) py pz zy
. predict double shat*, shares
. matrix list e(elast_price_nc)
. estat engel

. sysuse mex_bench, clear
. svyset
. easi w1 w2 w3, lnprices(lp1 lp2 lp3) lnexpenditure(lx) demographics(z1 z2)
    power(3) py vce(svy) stars saveres(tables.docx)
. easi, elasticities(market)
. easi w1 w2 w3, lnprices(lp1 lp2 lp3) lnexpenditure(lx) demographics(z1 z2)
    power(3) py vce(svy) hhsize(hhsize)

. easidiag w1 w2 w3 [pw=sweight], lnprices(lp1 lp2 lp3) lnexpenditure(lx)
    demographics(z1 z2) power(3) selvars(age isMale)
. easi w1 w2 w3, lnprices(lp1 lp2 lp3) lnexpenditure(lx) demographics(z1 z2)
    power(3) vce(svy) selvars(age isMale)
. predict double Ew*, shares
. easi w1 w2 w3, lnprices(lp1 lp2 lp3) lnexpenditure(lx) demographics(z1 z2)
    power(3) selvars(age isMale) vce(bootstrap, reps(500) seed(1) svy)
```

8.7 Speed

On *hixdata* (4847 observations, 9 goods, 576 parameters) the first command runs in 0.6 seconds in `compat` mode, which is the like-for-like comparison with the R package's 46 seconds on the same specification. That figure was measured on the same machine, twice: 45.8 seconds for the package's own code sourced under R 4.3.0 (`make_reference.R`), and 43 to 46 seconds over three runs of the compiled package under R 3.4.4 and `systemfit` 1.1.22 (`check_binary_r34.R`); the two R routes cost the same, and neither is slowed by the comparison code, which runs after the fit. The gap is not in the arithmetic of the estimator. Profiling the R call puts 96% of its time inside `systemfit`, in three places: stacking the eight equations into a $38\,776 \times 320$ sparse

system at every call (about a third), the GLS products $X'(\hat{\Sigma}^{-1} \otimes I)X$ taken on that stacked system rather than as $\hat{\Sigma}^{-1} \otimes A$ on a 72×72 block (another third), and the symmetry restrictions, which the package passes as character strings and `systemfit` re-parses with regular expressions at every call (a fifth) — all of it repeated over two passes of three to four iterations, the first pass being discarded (Appendix A, E2). `easi` never forms the stack, builds the restriction matrix once, and works on the $n \times k$ design.

Table 13: Running times, in seconds, on a laptop (Stata 19.5 SE, Windows 11). *hixdata*: the reference specification, 4,847 households, 9 goods, $R = 5$, all interactions, 576 coefficients. Mexican survey: 2,477 households, 3 goods, $R = 3$, `vce(svy)`.

<i>hixdata</i> , <code>compat</code> , no elasticity standard error	0.63
<i>hixdata</i> , default, no elasticity standard error	9.47
<i>hixdata</i> , default, with the elasticity standard errors	12.38
<i>hixdata</i> , default, every reporting table	12.55
<i>hixdata</i> , elasticities of the individuals	10.78
<i>hixdata</i> , conventional variance, no elasticity standard error	6.32
Mexican survey	0.06
Mexican survey, households that do not buy	1.27
the same, no elasticity standard error	0.35
the same, one replication of <code>vce(bootstrap)</code>	0.39

The default mode spends nine and a half seconds on the variance of the reference specification — the iterated $\hat{\Sigma}$, the generated-regressor term, the robust sandwich — and three more on the standard errors of the elasticities, which are the full influence functions of 216 means for the households and as many for the market. The correction for the households that do not buy takes about a second on the Mexican survey, most of it on the variance, whose Jacobians in the coefficients and in the probits are taken by central differences; a bootstrap of 500 replications of the whole procedure, which skips the analytic standard errors, takes three minutes.

8.8 Results on *hixdata*

The first command of the example, with `compensated demoelast checks` added, produces the following. The header and the expenditure elasticities are reproduced as Stata prints them; the $J \times J$ blocks, which Stata prints in one piece, are typeset as Tables 14 to 16. Prices are in logarithms normalised around the base period, so Φ is close to one (Section 4); `snames()` supplies the short good labels.

```

-----
EASI demand system                Number of obs   =      4,847
Goods                             = 9                Coef. per eq.   =       72
y power                           = 5                Coefficients    =      576
Interactions                       = py pz zy         Mode           =   corrected
    prices x                       : age hsex carown time tran
Elasticities                       = households (each household counts for its weight)
Variance                           = robust
Converged in 3 iteration(s), criterion = 3.093e-08 Execution time =    13.09 s
-----

```

Table 02: Expenditure elasticities (households), $d \ln q / d \ln x$

Good	Elasticity
foodh	0.5753
foodr	1.4349
rent	0.7308
oper	0.9949

furn		2.0083
cloth		1.5996
tranop		0.8373
recr		1.7134
pers		1.0697

Table 02-b: Expenditure elasticities (households), $d \ln q / d \ln x$ (standard errors)		

Good		Elasticity
-----+-----		
foodh		0.0209
foodr		0.0368
...		
Aggregation identities (exact by construction)		
Engel		$\sum_j w_j \eta_{x_j} - 1$ 2.097e-10
Cournot		$\max_k \sum_j w_j \eta_{x_j} + w_k$ 5.551e-17

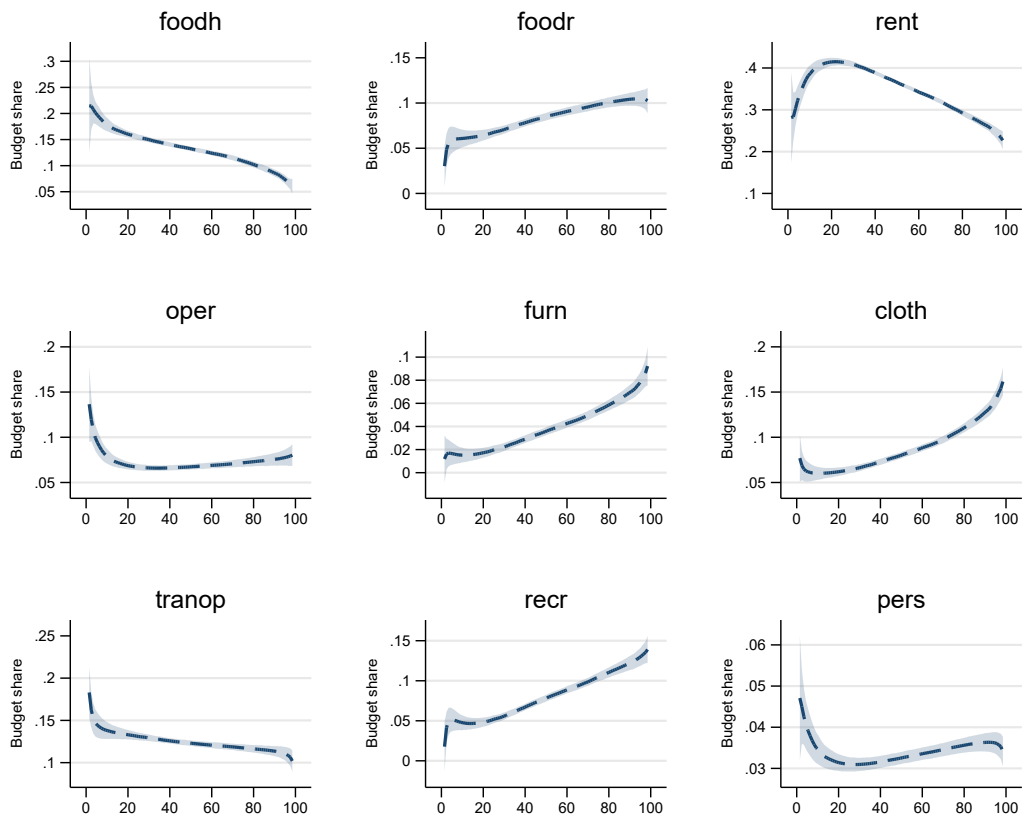
Food at home, rent and transport are necessities; food out, furnishings, clothing and recreation are luxuries, furnishings markedly so. Eight of the nine own-price elasticities in Table 14 are negative, and so are their compensated counterparts in Table 15; the difference between the two tables is the income effect of the Slutsky decomposition, the mean of $w_{hk}\eta_{hj}^x$ over the households, weighted as the elasticities are, largest for rent, the good with the largest share. The exception is personal care, the good recovered by adding up: +0.32 uncompensated and +0.37 compensated, with a standard error of 0.49 — an interval that covers -1 as comfortably as it covers zero. The sign is in the data, not in the code: the R package’s own formulas return +0.41 on the same cell. Two things explain the width. The reconstructed good’s price coefficients are minus the sum of the other eight, so its variance accumulates every covariance among them; and the price block is identified off 48 distinct price vectors (Section 7.1), with personal care the smallest share of the nine. A positive compensated own-price response violates the negativity of the Slutsky matrix, which EASI does not impose — symmetry is a linear restriction and is imposed, curvature is not — so the point estimate can sit on the wrong side of zero when the data do not pin it down; a household-level concavity check is the natural follow-up. If an economic mechanism is wanted behind the sign, it is not a Giffen one — the income effect of a normal good with a 2.8% share is 0.03, and the whole of +0.37 sits in the compensated response — but the quality effect of Deaton (1988): personal care is a composite, its price is a unit value, and when that price rises households move to cheaper varieties within the category, so the recorded *share* can rise while the quantity falls. The standard errors are widest for furnishings, the smallest share among the goods with strong price responses. The demographic elasticities in Table 16 are semi-elasticities of the shares: a car-owning household devotes 10.4 points less of its budget to transport operation, at a standard error of 0.2.

Figure 1 shows `estat engel` in its two forms. At the top, the exact Engel curves: demographics and prices held at their weighted means, the fixed point in (w, y) solved at each grid point, so that the curves are the model’s own functions of expenditure, with delta-method bands. At the bottom, the fitted shares as observed, smoothed by a local linear regression with a common bandwidth of 0.104, the rule of thumb for local polynomial regression; these also carry the covariation of demographics and prices with expenditure, which is why the two panels differ where households at the top of the distribution differ from the mean household in more than their expenditure.

8.9 Installation and compatibility

`easi` is twelve text files — `easi.ado`, `easi_p.ado`, `easi_estat.ado`, `easi.sthlp`, `easi.dlg`, `easidiag.ado`, `easidiag.sthlp`, `easi_examples.ado`, `_easi_tabstars.ado`, `_easi_pimpute.ado`, `sr_easi.ado`, `sr_easi_estat.ado` — and the two data sets of the ex-

Engel curves, EASI estimation



Percentiles of total expenditure

Fitted budget shares, at the mean of the demographics and of prices
tails trimmed at 1%; 95% confidence band

Engel curves, EASI estimation

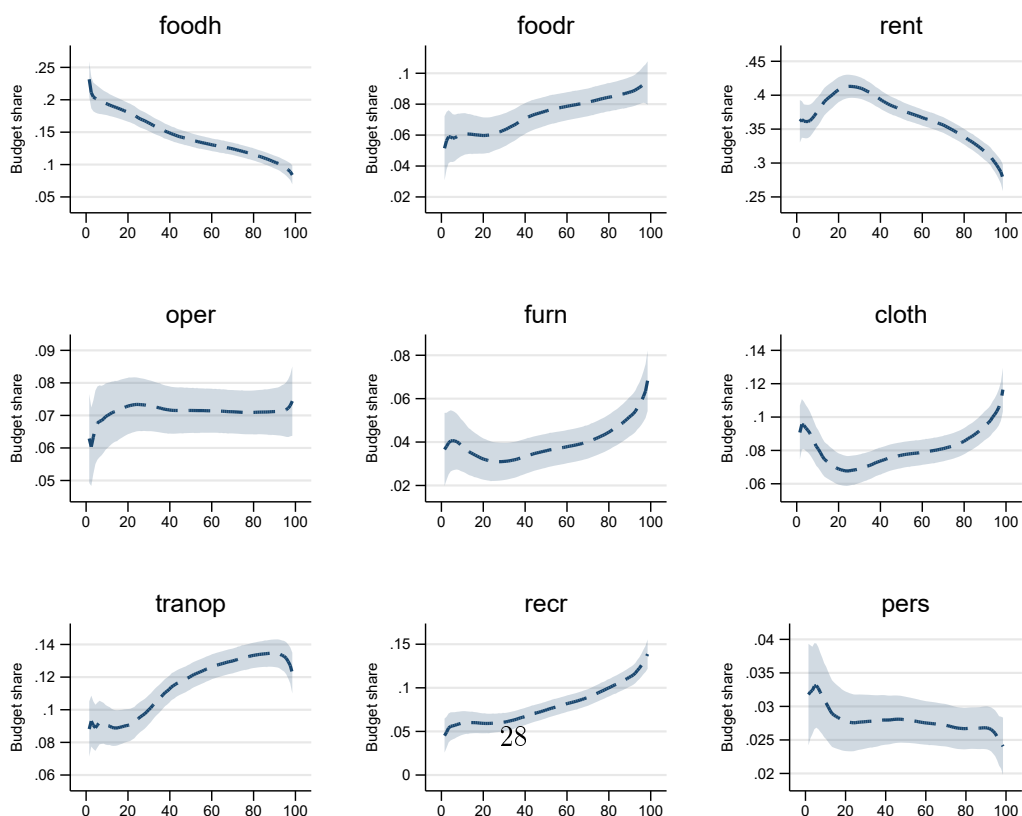


Table 14: Uncompensated (Marshallian) price elasticities, $\partial \ln q_j / \partial \ln p_k$, with delta-method standard errors in parentheses. *hixdata*, reference specification.

good	price of								
	foodh	foodr	rent	oper	furn	cloth	tranop	recr	pers
foodh	-0.838 (0.214)	0.113 (0.152)	0.078 (0.125)	0.039 (0.129)	0.605 (0.177)	-0.165 (0.133)	-0.054 (0.110)	-0.247 (0.153)	-0.107 (0.071)
foodr	0.099 (0.301)	-0.824 (0.361)	0.273 (0.229)	0.275 (0.229)	-0.834 (0.335)	-0.142 (0.283)	-0.376 (0.189)	0.329 (0.306)	-0.233 (0.136)
rent	0.000 (0.049)	0.111 (0.046)	-0.817 (0.054)	-0.023 (0.035)	0.081 (0.048)	-0.109 (0.042)	0.054 (0.040)	-0.041 (0.045)	0.012 (0.021)
oper	0.025 (0.264)	0.310 (0.236)	-0.199 (0.180)	-0.753 (0.276)	-0.067 (0.259)	-0.251 (0.234)	-0.255 (0.163)	0.508 (0.245)	-0.314 (0.121)
furn	2.026 (0.643)	-1.600 (0.622)	0.288 (0.441)	-0.188 (0.464)	-2.113 (1.069)	0.037 (0.612)	-0.926 (0.366)	-0.104 (0.803)	0.572 (0.375)
cloth	-0.432 (0.237)	-0.146 (0.256)	-0.809 (0.191)	-0.262 (0.205)	0.034 (0.298)	-0.912 (0.327)	0.487 (0.159)	0.449 (0.274)	-0.009 (0.154)
tranop	-0.108 (0.140)	-0.195 (0.121)	0.116 (0.127)	-0.145 (0.101)	-0.270 (0.127)	0.411 (0.113)	-0.743 (0.136)	0.125 (0.124)	-0.027 (0.054)
recr	-0.603 (0.280)	0.273 (0.282)	-0.553 (0.209)	0.410 (0.219)	-0.038 (0.399)	0.453 (0.280)	0.067 (0.178)	-1.661 (0.392)	-0.062 (0.149)
pers	-0.627 (0.373)	-0.587 (0.358)	0.032 (0.274)	-0.809 (0.310)	0.851 (0.533)	0.019 (0.449)	-0.139 (0.221)	-0.129 (0.427)	0.320 (0.492)

amples (`sysuse hixdata`, `sysuse mex_bench`), with nothing to compile, installed with

```
. net install easi, ///
    from("https://raw.githubusercontent.com/aabdd12/easi/v2.0.0")
```

from the repository <https://github.com/aabdd12/easi>, whose root carries `stata.toc` and `easi.pkg`; `net get easi` fetches the examples. The address installs the version this note documents; replacing `v2.0.0` by `main` installs the latest one. Stata 14.2 or later is required. The Mata code lives in `easi.ado`; the `predict`, `estat` and `easidiag` entry points are relays into it, because Mata functions defined in an ado-file are private to that file. `sr_easi` is retained as a wrapper: it accepts the old option spellings, accepts and ignores `rtool()`, and calls `easi` in `compat` mode, so existing do-files, saved dialog projects and menu entries continue to work and to give exactly the results they gave before.

9 Conclusion

`easi` estimates the EASI demand system within Stata alone, and is the only packaged implementation of the model currently maintained. Its quantity elasticities follow from differentiating the cost identity, (13)–(14), and satisfy homogeneity and the Slutsky decomposition; they are reported as means over the households of the elasticities of each household, for the household, the individual or the market, never for a reference household that no data describe. The residual covariance is iterated, so the estimator is invariant to the choice of dropped good; the variance carries the generated-regressor term, is heteroskedasticity-robust by default, and reads a stratified cluster design from `svyset`; the standard error of every elasticity comes from its full influence function, which treats the two means it is the ratio of as the estimators they are. The households that do not buy are handled by the two-step method of [Shonkwiler and Yen \(1999\)](#), with their missing prices imputed and their implicit utility completed, and with a diagnostic of the identification of the correction; `easidiag` finds, before estimation, what would make the estimation fail. The standard errors agree with the influence function computed by brute force, with a bootstrap of the whole procedure under simple random sampling, sampling weights and

Table 15: Compensated (Hicksian) price elasticities at constant utility, with standard errors. *hixdata*, reference specification.

good	price of								
	foodh	foodr	rent	oper	furn	cloth	tranop	recr	pers
foodh	-0.712 (0.214)	0.143 (0.152)	0.291 (0.124)	0.080 (0.129)	0.623 (0.177)	-0.125 (0.133)	0.003 (0.110)	-0.211 (0.153)	-0.092 (0.071)
foodr	0.283 (0.301)	-0.652 (0.361)	0.760 (0.229)	0.370 (0.229)	-0.782 (0.335)	-0.024 (0.282)	-0.206 (0.189)	0.446 (0.306)	-0.194 (0.136)
rent	0.116 (0.049)	0.152 (0.046)	-0.500 (0.054)	0.029 (0.035)	0.103 (0.048)	-0.060 (0.042)	0.126 (0.040)	0.004 (0.045)	0.031 (0.021)
oper	0.164 (0.264)	0.382 (0.237)	0.148 (0.178)	-0.658 (0.276)	-0.026 (0.259)	-0.170 (0.234)	-0.143 (0.163)	0.589 (0.245)	-0.285 (0.121)
furn	2.283 (0.644)	-1.450 (0.622)	0.949 (0.440)	-0.048 (0.464)	-1.962 (1.069)	0.222 (0.612)	-0.694 (0.366)	0.069 (0.803)	0.631 (0.375)
cloth	-0.224 (0.238)	-0.022 (0.256)	-0.271 (0.191)	-0.149 (0.205)	0.108 (0.298)	-0.734 (0.327)	0.669 (0.159)	0.581 (0.274)	0.042 (0.154)
tranop	0.004 (0.140)	-0.132 (0.121)	0.403 (0.127)	-0.089 (0.101)	-0.240 (0.127)	0.475 (0.113)	-0.604 (0.136)	0.190 (0.124)	-0.006 (0.054)
recr	-0.385 (0.280)	0.412 (0.282)	0.020 (0.208)	0.526 (0.219)	0.034 (0.399)	0.593 (0.280)	0.272 (0.178)	-1.453 (0.393)	-0.018 (0.149)
pers	-0.478 (0.373)	-0.514 (0.358)	0.407 (0.274)	-0.728 (0.309)	0.898 (0.533)	0.124 (0.449)	-0.024 (0.221)	-0.052 (0.428)	0.367 (0.492)

Table 16: Demographic elasticities $\partial w_j / \partial z_t$ (semi-elasticities of the shares), with standard errors. *hixdata*, reference specification.

demographic	good								
	foodh	foodr	rent	oper	furn	cloth	tranop	recr	pers
age	0.0012 (0.0001)	-0.0004 (0.0001)	0.0010 (0.0001)	-0.0002 (0.0001)	-0.0001 (0.0001)	-0.0004 (0.0001)	-0.0002 (0.0001)	-0.0009 (0.0001)	-0.0001 (0.0000)
hsex	-0.0117 (0.0020)	-0.0336 (0.0020)	0.0123 (0.0030)	0.0181 (0.0012)	0.0100 (0.0014)	0.0266 (0.0014)	-0.0129 (0.0017)	-0.0256 (0.0018)	0.0168 (0.0005)
carown	0.0181 (0.0022)	0.0096 (0.0022)	0.0346 (0.0033)	0.0091 (0.0015)	0.0078 (0.0016)	0.0117 (0.0016)	-0.1042 (0.0019)	0.0099 (0.0022)	0.0033 (0.0006)
time	0.0001 (0.0005)	-0.0018 (0.0006)	0.0027 (0.0004)	0.0007 (0.0004)	0.0002 (0.0007)	-0.0021 (0.0005)	-0.0006 (0.0003)	0.0006 (0.0006)	0.0001 (0.0003)
tran	0.0116 (0.0030)	-0.0127 (0.0028)	0.0026 (0.0043)	0.0087 (0.0021)	0.0037 (0.0023)	-0.0069 (0.0022)	-0.0051 (0.0026)	-0.0009 (0.0028)	-0.0009 (0.0008)

a survey design, and with the dispersion of the estimator in Monte Carlo experiments on data generated by the model, with and without non-buyers.

Where the reference code and the model disagree, the model decides. `compat` reproduces the R package to the eleventh decimal so that published results remain reproducible, and Appendix A records every departure with its measured effect.

A Relation to the R package `easi`

A.1 Status and reproduction

Hoareau et al. (2012) ported Pendakur’s Stata code to R and extended it with elasticities, Engel curves, equivalent incomes and simulations. CRAN archived the package on 22 August 2014 (“problems with vignette location and overly large PDF”, never addressed); version 0.21 remains in the CRAN archive and is what `sr_easi` called. `easi` began as a port of it: in `compat` mode it reproduces `easi` 0.21 to 1.8×10^{-11} on the coefficients and 1.3×10^{-12} on their standard errors,

on the 320-parameter specification of the package vignette — the same data columns, the same options and the same `elastic()` calls as the vignette’s own example. The reference output was produced twice. First by running the package’s own code under R 4.3.0 and `systemfit` 1.1.30, its functions sourced from the installed lazy-load database (the compiled package does not load under R 4) with one `formula()` patch that does not touch the estimation. Then, as a check on that route, by running the *compiled* package itself, `easi_0.21.zip` unmodified, under R 3.4.4 — the R that `sr_easi` shipped with — and `systemfit` 1.1.22, the dependencies being taken from a 2018 snapshot of the Posit package manager since CRAN no longer serves binaries for R 3.4. The two agree to 2.6×10^{-9} on the coefficient table and to between 10^{-9} and 10^{-15} on the six elasticity and standard-error tables (`replication/R_reference/check_binary_r34.R`). Every figure below that names a defect of the package is either visible in that output — the compiled package stores `interpz = 1` when `1:5` was requested, fits 40 coefficients per equation with `z1` as the only demographic crossed with prices, and returns -1.019 as the standard error of the dropped good — reproduced by a direct R evaluation (the `ifelse` and `round` behaviours), or implied by the reproduction itself, since `compat` could not match to eleven decimals without emulating the behaviour in question. The one exception is `engel()`, item (L8), which was read but not run. The differences are therefore established against a verified baseline, not inferred, and each is reproduced under `compat` and, individually, under the developer option `legacy(namelist)` with names `interpz instr quant ez lastse elast elastse signal condse`, which is how the effects in this appendix were measured. Agreement with the package was treated as a lock, not a target: it proves the port is understood, and where the package and the model disagree, the model decides.

A.2 Summary

Table 17 lists, in order of importance for what a user publishes, what the package does and what `easi` does.

A.3 Estimation

- (E1) **interpz never worked.** The package collapses the vector of demographics to be interacted with prices to its first element, through `interpz <- ifelse(length(interpz) > 1, interpz, 1:nsoc)`; `ifelse()` is vectorised and returns a result the length of its test. Requesting `interpz=1:5` therefore interacts prices with the first demographic only, and `interpz=c(3)` selects the first, not the third. `sr_easi` always passed `interpz=c(1:nitemsd)`, so `inpz(1)` only ever crossed prices with the first demographic. Effect: on the vignette specification the system grows from 320 to 576 parameters and price elasticities move by up to 2.0.
- (E2) **The refined instrument is discarded.** The package rebuilds $\tilde{y}^*$ at the end of every first-pass iteration but never refreshes the powers $(\tilde{y}^*)^r$, and then resets its data frame at the start of the second pass, discarding the interactions as well. The reported estimates are therefore obtained with the raw $\tilde{y}$, and the first pass — half the run time — has no effect beyond shifting the convergence bookkeeping by one iteration. Effect: 3.5×10^{-4} on the expenditure elasticities.
- (E3) **A mistranslated rounding.** Pendakur writes `int(1000000*pAp+0.5)/1000000`, ordinary rounding to six decimals: with $u = 10^6x$, $\lfloor u + \frac{1}{2} \rfloor$ moves up only when the fractional part of u is at least one half. The port writes `round(1e6*pAp+0.5)/1e6`, and `round(u + 0.5)` is $\lfloor u \rfloor + 1$ whenever the fractional part of u is positive: the expression is a ceiling half of the time. Run in R 4.3.0 on 10^5 uniform draws, the two differ on 50.1% of them, always by one unit, an average shift of 5×10^{-7} on $p'Ap$ and $p'Bp$. Effect on the elasticities: 2×10^{-7} .

Table 17: Package R `easi` 0.21 and `easi`, by importance.

item	R package	<code>easi</code>	effect / validation
quantity elasticities	y differentiated at fixed shares; partial $p'Bp$	cost identity, (12)	FD 1.9×10^{-2} vs 10^{-8} ; homogeneity 6×10^{-3} vs 6×10^{-9}
elasticity std. errors	median of pointwise semi-elasticity SE $\div$ mean share	full influence function, analytic Jacobian	$\times 14$ on dropped good $\rightarrow$ 0.99 / 1.01; brute force within 0.16%; Monte Carlo 0.96–1.03
survey design	none	<code>vce(svy)</code> , FPC	design bootstrap 1.02; ignoring clusters 1.33 max
sampling weights	none	<code>aw pw iw</code>	bootstrap 1.05 / 0.99 / 1.01
coefficient covariance	homoskedastic, given y	robust + generated regressor	1.23 (max 1.69) $\rightarrow$ 1.007
elasticity as a ratio types of elasticities	$\bar{w}$ fixed households, product of mean shares in the compensated	full influence function households, individuals, market; mean of products	+18% $\rightarrow$ within 5% compensated ± 0.07
non-buyers	none	<code>pimpute()</code> , <code>selection</code>	Monte Carlo 0.93–1.05; brute force within 0.5%
Σ	one step	iterated	0.20 $\rightarrow 10^{-7}$ on dropped good
demographic elasticities	accumulator and index errors	corrected	all cells; SE cell-by-cell 10^{-12}
<code>interpz</code>	collapsed to first element	as requested	320 $\rightarrow$ 576 parameters; ± 2.0
pre-estimation diagnostic instruments	none refined $\tilde{y}^*$ discarded	<code>easidiag</code> refreshed	predicted iterations exact 3.5×10^{-4} ; time $\div 2$
Engel curves	Stone term, price block, band errors	<code>estat engel</code> rewritten	—
<code>predict, e()</code>	none	complete	fixed point 6.7×10^{-16}
rounding of $p'Ap$, $p'Bp$	<code>round(u+.5)</code> , a ceiling half the time	Pendakur's <code>int(u+.5)</code>	2×10^{-7}
equivalent income, concavity, simulations	present	not ported	rest on the partial $p'Bp$
speed	46 s	0.6 s (<code>compat</code>), 12.4 s	—

(E4) **Weights were lost.** Pendakur's code carries [`aweight=obs_weight`] on every `reg3`. The R package has no weighting at all.

(E5) **The covariance is homoskedastic and conditional on y .** The package reports `systemfit`'s conventional 3SLS covariance, formed once from the 2SLS residuals and taken given the generated regressor y ; Pendakur's code notes, twice, that “reported standard errors are wrong for iterated estimates”. Section 5.4 measures the consequence.

A.4 Elasticities

(L1) **$\partial y / \partial \ell_k$ is taken with the shares held fixed.** The package differentiates (4) rather than the cost identity. Writing $y = u/v$ with u the numerator and v the denominator of (4), and differentiating *totally*,

$$\frac{\partial u}{\partial \ell_k} = -w_k - \sum_j \frac{dw_j}{d\ell_k} \ell_j + U_k, \quad U_k \equiv \sum_d \sum_t a^{kdt} z_t \ell_d, \quad \frac{\partial v}{\partial \ell_k} = -G_k,$$

with $G_k \equiv \sum_d b^{kd} \ell_d$. Substituting $dw_j/d\ell_k = \Psi_{jk} + \aleph_j dy/d\ell_k$ and using symmetry to get $\sum_j \Psi_{jk} \ell_j = U_k + y G_k$, the terms $U_k + y G_k$ cancel and one recovers (12). The package drops the feedback term $\sum_j (dw_j/d\ell_k) \ell_j$ — it treats the shares as constants — so those terms

survive spuriously and the denominator is v instead of Φ : it uses $B = (-w_k + U_k + yG_k)/v$ where (12) gives $-w_k/\Phi$.

- (L2) $\partial y/\partial \ln x = 1$ is **assumed**, where (12) gives $1/\Phi$.
- (L3) **The code does not implement its own derivation.** In the expression for B , U and G carry the opposite sign, and both are indexed by the equation rather than by the price — yet $\partial y/\partial \ell_k$ cannot depend on which share is being differentiated. Repairing the code to match the vignette would therefore not be enough.
- (L4) **$p'Bp$ is a partial sum over raw prices.** The quantity needed is $\sum_{j,k \leq J} b^{jk} \ell_j \ell_k$, equivalently $\sum_{j,k \leq J-1} B_{jk} \text{np}_j \text{np}_k$; the package sums only over the $J-1$ estimated goods but with raw log prices. This is what breaks homogeneity. The same defect affects the quantities used by `equiv.income` and `concavity`, which is why they were not ported.
- (L5) **Demographic elasticities.** The running sum over price terms is initialised outside the loop over demographics, so effects accumulate from one variable to the next; and the coefficient array is indexed one position short, so the constant block is read instead of the first demographic.
- (L6) **Standard error of the dropped good.** It is returned with a minus sign, and two closing cells omit the squares in a root-sum-of-squares.
- (L7) **Standard errors of the elasticities.** The package reports, for each elasticity, the *median across households* of the pointwise delta-method standard error of the corresponding *semi*-elasticity, divided by the mean budget share. That is not the standard error of anything it reports. Against the bootstrap of Section 5.7 it is 40% too large on the expenditure elasticities — the median of pointwise norms standing in for the norm of the average gradient — and fourteen times too large on the dropped good, where a root-sum-of-squares across equations throws away the negative covariances that adding up creates. It is right on the own-price elasticities only because two errors of opposite sign, the median of norms and the omitted $B(C + D + G)$ term, cancel.
- (L8) **Engel curves.** `engel()` overwrites the estimator's y with a recomputation whose Stone term omits the last good; it puts the entire price block behind the `pz.inter` switch, so the price contribution disappears when there are no price–demographic interactions; and it builds the confidence band from raw rather than normalised log prices.

A.5 Verification of the elasticity formulas

Three independent checks, none relying on the others.

Finite differences. Equations (4) and (5) define a fixed point. Perturbing ℓ_k (or $\ln x$), re-solving for (w, y) and reading off $dw_j/d\ell_k$ gives the elasticity by construction, with no derivation to get wrong. The fixed point reproduces the observed shares to 6.7×10^{-16} , confirming that the reconstructed J -good representation is equivalent to the estimated $(J-1)$ -equation system.

Homogeneity. Multiplying every price and total expenditure by the same factor cannot change any elasticity. The model satisfies this: the implicit utility index and the fitted shares are invariant to 5×10^{-8} , which is the slack in $|\sum_j w_j - 1|$ in the data itself. The price elasticities reported by the package are *not* invariant — they move by 6.0×10^{-3} — while the corrected ones move by 5.9×10^{-9} . No correct formula can fail this test, and the test needs no external reference.

Table 18: Maximum absolute deviation from the finite-difference benchmark. *hixdata*, $J = 9$, $R = 5$, all interactions.

	expenditure elasticities	price elasticities
(13)–(14), closed form	2.5×10^{-11}	3.9×10^{-11}
<code>easi</code>	7.1×10^{-9}	1.8×10^{-8}
<code>easi</code> , <code>compat</code> (= R package)	1.9×10^{-2}	2.6×10^{-2}

Slutsky. The corrected expressions satisfy $\eta_j^k = (\Gamma_{jk} - \mathbf{1}\{j = k\}) - w_k \eta_j^x$ using the package’s own compensated matrix Γ , which is correctly computed. The package’s expressions do not: it is internally inconsistent.

A.6 How much the formulas matter

$\Phi = 1$ exactly when all log prices are zero, that is, at the usual base-period normalisation; any formula that omits Φ is exact at that point and only there. The error is proportional to $1 - 1/\Phi$, and since Φ is invariant to a common price shift, it departs from 1 with the *dispersion* of relative log prices, not with their level. On *hixdata*, where prices are normalised around the base period, Φ ranges from 0.948 to 1.143 with mean 1.015, and the elasticities are off by 1–2%. The error is systematic rather than random, so it does not average out, and it is not bounded: the wider the relative price variation — long panels, regional comparisons, differentiated inflation — the larger it becomes.

B What the test suite verifies

The repository’s `replication/` folder carries, under descriptive names and with a `master.do`, the scripts that produce every table and figure of this note from the two data sets in `examples/`; its README maps each script to what it reproduces. Most of them are the module’s test suite: each locks one property and fails on any departure, so a clean run is itself the statement that the numbers reproduce. In order:

- 0–2 the reference R output is frozen, and confirmed by the compiled package under R 3.4.4 to 2.6×10^{-9} ; `compat` reproduces it to 1.8×10^{-11} (coefficients) and 1.3×10^{-12} (standard errors), elasticities and their standard errors included;
- 3 each `legacy()` switch is measured one at a time;
- 4 weights (`aw` `pw` `iw`, `fw` refused), `vce(robust)`, `vce(cluster)`, `predict` in and out of sample, the help file, the dialog, the `sr_easi` wrapper, the deliverable files and the `net` manifest;
- 5 the elasticity formulas against the finite-difference fixed point, homogeneity and Slutsky;
- 6 `estat engel`;
- 7–8 the coefficient and elasticity standard errors against the bootstrap (Tables 3, 6);
- 9 the delta method and invariance to the dropped good (Table 1);
- 10–11 the generated-regressor Jacobian against a numerical Jacobian of an independently reimplemented moment vector (10^{-9}), and the analytic elasticity Jacobian against forward differences;
- 12 the elasticity tables: orientation, $\hat{\Gamma} - I$, Engel and Cournot aggregation on the user’s data;

- 13 **easidiag**: no estimation, near-dependency (not only exact) detected and named, iterations predicted;
- 14–15 **pweight** standard errors against a weighted bootstrap; **vce(svy)** with FPC against the design bootstrap, elasticities included (Tables 4, 5);
- 16 the analytic elasticity Jacobian against finite differences on six specifications and four families, and the demographic standard errors cell by cell against $\hat{V}$;
- 17–19 the tables with stars and their export to five formats; **fweights** refused; the examples of the help file, in the command window, as do-files and in the dialog box, the data in memory kept;
- 20 the three types of elasticities: the market recomputed from outside on **e(b)** with the weights ωx , the individuals equal to the households weighted by ωn (every result of **e()**), the total Jacobian of the elasticities, y term included, against central differences;
- 21–27 the households that do not buy: **pimpute()** against an oracle; the probits against Stata's **probit**; the general GMM engine equal to the 3SLS without selection; the estimator against **ivregress 2sls**, **predict** and the completion of y against closed forms; **vce(bootstrap)**; the selection section of **easidiag** against $1/(1-R^2)$; the elasticities of the expected demand against finite differences.

The validation scripts of Tables 2, 7, 8, 9 and 10 are not tests but measurements, and **replication/master.do** runs them too: the brute force and the Monte Carlo experiments take about four hours.

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